Advanced Linear Algebra II

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Introduction

Linear algebra begins with concrete objects: systems of linear equations, matrices, determinants, vector spaces, and linear maps. After these foundations are established, the subject gradually reveals a more structural character. Questions about matrices become questions about linear transformations; changes of coordinates lead to similarity and congruence; eigenvalues lead to canonical forms; and inner products introduce geometric structure into vector spaces.
The first two volumes of these notes remain within this finite-dimensional undergraduate framework.
The first volume develops the basic language and machinery of higher algebra:

$$\text{sets and maps} \longrightarrow \text{linear systems and matrices} \longrightarrow \text{vector spaces} \longrightarrow \text{linear maps},$$

followed by determinants, polynomials, eigenvalues, and diagonalization.
The second volume develops finer finite-dimensional structure:

$$\begin{aligned} \text{Jordan normal form} &\longrightarrow \text{inner product spaces} \\ &\longrightarrow \text{adjoints and normal operators} \\ &\longrightarrow \text{spectral theorem}, \end{aligned}$$

and then develops bilinear forms, quadratic forms, singular value decomposition, and the principal matrix factorizations.
Together, those two volumes are intended to form a coherent account of the material that belongs to an undergraduate sequence in advanced linear algebra.
This third volume has a different purpose.

0.1 Beyond the Boundary of Linear Algebra

An honors course in advanced linear algebra naturally reaches beyond the usual undergraduate syllabus. Once vector spaces and linear maps are viewed structurally, several constructions appear that no longer belong exclusively to linear algebra.
A vector space, for example, may be regarded as a special case of a module. The familiar quotient space

$$V/W$$

has an analogue for modules. Bilinear maps suggest the construction of tensor products. Direct products may be replaced by more complicated products determined by an action. Exact sequences provide a compact language for describing kernels, images, quotients, and extensions.
At the same time, several additional algebraic structures begin to appear:

$$\text{groups}, \qquad \text{rings}, \qquad \text{modules}, \qquad \text{algebras},$$

together with interactions among them.
These ideas form bridges from linear algebra to later subjects rather than a continuation of elementary matrix theory.
For this reason, they have deliberately been removed from the first two volumes and collected separately here.

0.2 The Purpose of This Volume

This volume should not be read in exactly the same way as the first two.
The first two volumes are intended to represent material that should eventually become part of a working undergraduate knowledge of linear algebra. Definitions and theorems there are included with the expectation that their meaning, motivation, and use can be understood within the ordinary development of the subject.
The present volume is instead an archive of advanced algebraic topics encountered during the honors course.
Its purpose is

$$\boxed{\text{preservation before mastery.}}$$

Some of the material collected here may require substantially more algebraic maturity than the linear algebra from which it arose. The fact that a definition or theorem appears in these notes should therefore not be interpreted as a claim that it has already been fully understood.
The objective is more modest: to preserve the mathematical objects, definitions, statements, constructions, and relationships appearing in the course in a form that can be located and studied again when they become relevant.
In this sense, the volume is intentionally closer to a structured reference than to a self-contained textbook.

0.3 From Vector Spaces to Modules

One of the first steps beyond ordinary linear algebra is to weaken the role played by the scalar field.
A vector space V allows multiplication

$$K \times V \to V$$

by elements of a field K.
If the field is replaced by a ring R, one obtains the notion of an R-module.
Schematically,

$$\boxed{\text{vector space} = \text{module over a field.}}$$

This apparently small change has major consequences.
Over a field, every nonzero scalar is invertible. This property lies behind many familiar results of linear algebra. For example, finite-dimensional vector spaces possess bases, subspaces admit complements, and dimension behaves particularly well.
For modules over general rings, these properties can fail.
Thus module theory preserves the basic language of addition and scalar multiplication while removing one of the strongest assumptions in linear algebra.
Submodules and quotient modules then extend the familiar constructions

$$W \subseteq V, \qquad V/W$$

from vector spaces to a much broader algebraic setting.

0.4 Tensor Products and Multilinear Structure

Ordinary linear algebra is built around linear maps

$$T : V \to W.$$

However, many natural constructions depend on several variables. A typical example is a bilinear map

$$B : V \times W \to U.$$

The tensor product provides a way to convert such multilinear information into linear information.
At a formal level, the central principle is

$$\boxed{\text{bilinear maps on } V \times W \longleftrightarrow \text{linear maps on } V \otimes W.}$$

Thus tensor products are not introduced merely as another operation on vector spaces. They provide a universal object through which bilinear maps factor.
This construction later appears throughout algebra, geometry, analysis, and mathematical physics. In the present volume, however, the emphasis remains on the algebraic definitions and constructions encountered in the course.

0.5 Extensions and Product Structures

Direct sums and direct products describe situations in which two algebraic objects coexist without interacting strongly.
More general constructions allow one object to act on another.
This leads, for example, to semidirect products. Instead of treating two groups independently, a semidirect product records an action of one group on another and incorporates that action into the multiplication law.
Schematically,

$$\boxed{\text{direct product} + \text{nontrivial action} \longrightarrow \text{semidirect product.}}$$

Such constructions illustrate a broader theme of advanced algebra: new structures are often created not merely by combining objects, but by recording how those objects interact.
Exact sequences provide another language for such relationships. A sequence such as

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

encodes the idea that A occurs inside B, while C is obtained as an appropriate quotient. This language becomes increasingly important in later algebra.

0.6 Further Algebraic Structures

The boundary between advanced linear algebra and later algebraic subjects is not sharp. Once modules, tensor products, and algebraic actions are introduced, one naturally encounters structures that belong more properly to abstract algebra and its descendants.
Among them are Lie algebras, which replace an associative product by a bracket

$$[x, y],$$

and commutative algebra, which studies commutative rings, their ideals, modules, and related constructions at a substantially deeper level.
These subjects are too large to be developed systematically in a volume originating from an advanced linear algebra course.
Accordingly, when such topics occur here, they are included only to the extent needed to preserve the material encountered in the course and to indicate where it belongs in the larger algebraic landscape.
They should be regarded as entrances to later subjects rather than as complete treatments.

0.7 Organization of the Three-Volume Sequence

The three volumes may therefore be understood as having three different levels.
The first volume develops the foundations:

$$\boxed{\text{coordinates} \longrightarrow \text{vector spaces} \longrightarrow \text{linear transformations} \longrightarrow \text{eigenvalue theory.}}$$

The second volume develops the structural and geometric theory:

$$\boxed{\text{canonical forms} \longrightarrow \text{inner product geometry} \longrightarrow \text{spectral theory} \longrightarrow \text{matrix decompositions.}}$$

The present volume records the transition beyond ordinary linear algebra:

$$\boxed{\text{vector spaces} \longrightarrow \text{modules} \longrightarrow \text{tensor constructions} \longrightarrow \text{extensions and further algebraic structures.}}$$

The boundary is intentional.
Material is not placed in this volume merely because it is difficult. Jordan normal form, spectral theory, quadratic forms, and singular value decomposition can all be technically demanding, but they remain central parts of undergraduate linear algebra and therefore belong to the second volume.
By contrast, modules and tensor products alter the algebraic language itself and point directly toward later courses. They therefore belong here even when their elementary definitions are short.

0.8 How These Notes Should Be Used

This volume is best treated as a map for future study.
When a construction reappears in abstract algebra, representation theory, geometry, topology, or another advanced course, the relevant chapter can be revisited with greater mathematical context.
Accordingly, the notes emphasize

$$\text{definitions,} \qquad \text{basic constructions,} \qquad \text{principal statements,}$$

and

$$\text{relationships among structures.}$$

They do not attempt to disguise gaps in understanding by replacing them with excessive exposition. Nor do they attempt to develop every topic to the level of a specialized textbook.
The guiding principle is instead:

$$\boxed{\text{record accurately, organize clearly, and return when needed.}}$$

This distinguishes *Algebraic Frontiers* from the first two volumes. It is not intended to certify mastery of the material it contains. It preserves the frontier reached by the course so that unfamiliar ideas are not lost merely because they were encountered before the background required to understand them fully had been developed.
The chapters that follow therefore begin where ordinary finite-dimensional linear algebra ends and collect the algebraic structures that point toward the next stage of study.

Chapter 1

Modules and Quotient Modules

1.1 From Vector Spaces to Modules

Let K be a field. A vector space V over K consists of an abelian group under addition together with scalar multiplication

$$K \times V \to V$$

satisfying the familiar distributive and associative laws.
The definition of a module is obtained by replacing the field K with a ring R.

Definition 1.1.1. Let R be a ring. A left R-module is an abelian group $(M, +)$ together with a scalar multiplication

$$R \times M \to M, \qquad (r, m) \mapsto rm,$$

such that for all $r, s \in R$ and $m, n \in M$,

$$r(m + n) = rm + rn,$$

$$(r + s)m = rm + sm,$$

and

$$(rs)m = r(sm).$$

If R has an identity 1_R, we usually also require

$$1_R m = m.$$

Thus a vector space is simply a module whose scalar ring happens to be a field:

$$\boxed{\text{vector spaces} \subset \text{modules.}}$$

The change from fields to rings appears small, but it removes one of the strongest assumptions used throughout linear algebra: a nonzero scalar in a general ring need not be invertible.
This is the source of many differences between module theory and ordinary linear algebra.

1.2 Basic Examples of Modules

Example 1.2.1. Every vector space over a field K is a K-module.

Thus

$$\mathbb{R}^n, \qquad \mathbb{C}^n, \qquad K[x],$$

and spaces of matrices or functions are all examples of modules when viewed over their scalar fields.

Example 1.2.2. Every abelian group G is naturally a $\mathbb{Z}$-module.

For

$$n \in \mathbb{Z}$$

and

$$g \in G,$$

define

$$ng = \underbrace{g + \cdots + g}_{n \text{ times}}$$

when $n > 0$, with the usual extensions to $n = 0$ and $n < 0$.
Hence

$$\boxed{\mathbb{Z}\text{-modules} = \text{abelian groups.}}$$

This example is important because it shows that module theory simultaneously generalizes linear algebra and the theory of abelian groups.

Example 1.2.3. Every ring R is naturally a left R-module over itself, with scalar multiplication given by ring multiplication.

We denote this module simply by

$$_R R.$$

Example 1.2.4. The abelian group

$$\mathbb{Z}/n\mathbb{Z}$$

is a $\mathbb{Z}$-module.

This module already behaves differently from a vector space. For example, in

$$\mathbb{Z}/6\mathbb{Z},$$

the nonzero scalar $2 \in \mathbb{Z}$ satisfies

$$2 \cdot \overline{3} = \overline{0}.$$

Such phenomena cannot occur in a vector space when the scalar is nonzero.

1.3 Left and Right Modules

When R is not commutative, the side on which scalars act matters.
A left R-module has multiplication

$$R \times M \to M,$$

written

$$(r, m) \mapsto rm.$$

A right R-module has multiplication

$$M \times R \to M,$$

written

$$(m, r) \mapsto mr.$$

The associativity conditions are respectively

$$(rs)m = r(sm)$$

and

$$m(rs) = (mr)s.$$

If R is commutative, left and right modules can usually be identified by setting

$$mr = rm.$$

For most elementary examples in this chapter, the scalar ring will be commutative, so this distinction will not cause difficulty.
It becomes important later when tensor products over noncommutative rings are considered.

1.4 Submodules

The analogue of a vector subspace is a submodule.

Definition 1.4.1. Let M be an R-module. A subset $N \subseteq M$ is an R-submodule if

$$N$$

is an additive subgroup of M and

$$rn \in N$$

for every $r \in R$ and $n \in N$.

We write

$$N \leq M$$

or simply

$$N \subseteq M.$$

Equivalently, a nonempty subset $N \subseteq M$ is a submodule if

$$x - y \in N$$

and

$$rx \in N$$

for every $x, y \in N$ and $r \in R$.

Example 1.4.2. For the $\mathbb{Z}$-module $\mathbb{Z}$, every submodule has the form

$$n\mathbb{Z}$$

for some integer $n \geq 0$.

Thus the submodules of the regular $\mathbb{Z}$-module are precisely the ideals of $\mathbb{Z}$.
More generally, if a ring R is regarded as a left module over itself, its submodules are precisely its left ideals.
Therefore

ideals are special cases of submodules.

1.5 Generated Submodules

Let $S \subseteq M$.

Definition 1.5.1. The submodule generated by S, denoted by

$$\langle S \rangle_R,$$

is the smallest submodule of M containing S.

Explicitly,

$$\boxed{\langle S \rangle_R = \left\{ \sum_{i=1}^{k} r_i s_i : r_i \in R,\ s_i \in S,\ k \in \mathbb{N} \right\}.}$$

If

$$S = \{m_1, \ldots, m_n\},$$

we write

$$\langle m_1, \ldots, m_n \rangle_R.$$

Then

$$\langle m_1, \ldots, m_n \rangle_R = Rm_1 + \cdots + Rm_n.$$

This is the module analogue of the span of vectors.

1.6 Finitely Generated Modules

Definition 1.6.1. An R-module M is finitely generated if there exist

$$m_1, \dots, m_n \in M$$

such that

$$M = Rm_1 + \cdots + Rm_n.$$

We then say that

$$m_1, \dots, m_n$$

generate M.
This resembles a spanning set in linear algebra, but an important difference appears immediately.
For vector spaces, a finite spanning set can always be reduced to a basis.
For general modules, a generating set need not contain anything resembling a vector-space basis.
For example,

$$\mathbb{Z}/2\mathbb{Z}$$

is generated as a $\mathbb{Z}$-module by

$$\overline{1},$$

but it cannot have a basis over $\mathbb{Z}$ in the vector-space sense, because

$$2\overline{1} = 0.$$

1.7 Linear Independence over a Ring

The familiar definition of linear independence can still be stated.
Elements

$$m_1, \dots, m_n \in M$$

are linearly independent over R if

$$r_1 m_1 + \cdots + r_n m_n = 0$$

implies

$$r_1 = \cdots = r_n = 0.$$

However, the behavior of linear independence over a general ring is less rigid than over a field.
In a vector space, the theory of bases gives

$$\text{spanning} + \text{linear independence} \longrightarrow \text{dimension theory}.$$

For general modules, there may be no basis at all.
This motivates the special class of free modules.

1.8 Free Modules

Definition 1.8.1. An R-module M is free if there exists a subset

$$\mathcal{B} = \{e_i\}_{i \in I}$$

such that every element $m \in M$ can be written uniquely as a finite linear combination

$$m = \sum_{i \in I} r_i e_i,$$

where all but finitely many r_i are zero.

The set $\mathcal{B}$ is called a basis of M.
Thus free modules are the modules that most closely resemble vector spaces.
The standard example is

$$R^n.$$

Its standard basis is

$$e_1, \ldots, e_n,$$

where

$$e_i = (0, \ldots, 0, 1, 0, \ldots, 0).$$

Hence

$$R^n = Re_1 \oplus \cdots \oplus Re_n.$$

Example 1.8.2. The $\mathbb{Z}$-module

$$\mathbb{Z}^n$$

is free of rank n.

Example 1.8.3. The $\mathbb{Z}$-module

$$\mathbb{Z}/n\mathbb{Z}$$

is not free when $n > 1$.

Indeed,

$$n\overline{1} = 0,$$

although

$$n \neq 0$$

in $\mathbb{Z}$.

1.9 Rank of a Free Module

For vector spaces, any two bases have the same cardinality.
A similar statement holds for free modules over many rings of interest, including commutative rings with identity.
For a finite free module

$$M \cong R^n,$$

we call n the rank of M:

$$\boxed{\operatorname{rank}_R(M) = n.}$$

This resembles vector-space dimension, but the analogy should not be extended carelessly.

A general module need not be free and therefore need not possess a rank defined in this elementary sense.
Thus

$$\boxed{\text{dimension}}$$

is universal for finite-dimensional vector spaces, while

$$\boxed{\text{rank of a free module}}$$

applies only after freeness has been established.

1.10 Module Homomorphisms

The analogue of a linear transformation is a module homomorphism.

Definition 1.10.1. Let M and N be R-modules. A map

$$f : M \to N$$

is an R-module homomorphism if

$$f(x + y) = f(x) + f(y)$$

and

$$f(rx) = rf(x)$$

for all $x, y \in M$ and $r \in R$.

We write

$$\mathrm{Hom}_R(M, N)$$

for the collection of all R-module homomorphisms from M to N.
Thus

$$\boxed{\text{linear maps} = \text{module homomorphisms over a field.}}$$

As in linear algebra, define

$$\ker f = \{m \in M : f(m) = 0\}$$

and

$$\mathrm{Im}\, f = \{f(m) : m \in M\}.$$

Both are submodules.

1.11 Isomorphisms of Modules

Definition 1.11.1. An R-module homomorphism

$$f : M \to N$$

is an isomorphism if it is bijective.

If such an isomorphism exists, we write

$$M \cong_R N.$$

For example,

$$R^m \cong_R R^n$$

only under appropriate conditions on the ring and the integers m, n.
For ordinary finite-dimensional vector spaces,

$$K^m \cong_K K^n \iff m = n.$$

Module theory asks how much of this familiar classification survives after the scalar field is replaced by a ring.

1.12 Sums of Submodules

Let $N_1, N_2 \leq M$.
Their sum is

$$N_1 + N_2 = \{n_1 + n_2 : n_1 \in N_1,\ n_2 \in N_2\}.$$

This is again a submodule of M.
More generally,

$$N_1 + \cdots + N_k$$

is the submodule consisting of all sums

$$n_1 + \cdots + n_k$$

with

$$n_i \in N_i.$$

Their intersection

$$N_1 \cap N_2$$

is also a submodule.
Thus the familiar operations on vector subspaces extend naturally to modules.

1.13 Direct Sums of Modules

Definition 1.13.1. Let $M_1, \ldots, M_n$ be R-modules. Their direct sum is

$$M_1 \oplus \cdots \oplus M_n$$

with componentwise addition and scalar multiplication.

If $N_1, N_2 \leq M$, then

$$M = N_1 \oplus N_2$$

means

$$M = N_1 + N_2$$

and

$$N_1 \cap N_2 = \{0\}.$$

Equivalently, every $m \in M$ has a unique representation

$$m = n_1 + n_2$$

with

$$n_i \in N_i.$$

This is formally identical to the direct-sum construction for vector spaces.
However, an important difference is that a submodule need not admit a complement.

1.14 Submodules Need Not Have Complements

For a finite-dimensional vector space V, every subspace $W \subseteq V$ has a complementary subspace U such that

$$V = W \oplus U.$$

This fails for modules.
Consider the $\mathbb{Z}$-module

$$M = \mathbb{Z}$$

and its submodule

$$N = 2\mathbb{Z}.$$

Suppose there were a submodule L such that

$$\mathbb{Z} = 2\mathbb{Z} \oplus L.$$

Every submodule of $\mathbb{Z}$ has the form

$$m\mathbb{Z}.$$

But no such submodule can provide a direct complement of $2\mathbb{Z}$.
Thus

$$\boxed{\text{submodules need not split off as direct summands.}}$$

This is one of the first major points at which module theory departs from ordinary finite-dimensional linear algebra.

1.15 Cosets of a Submodule

Let

$$N \leq M.$$

For $m \in M$, define the coset

$$m + N = \{m + n : n \in N\}.$$

Two elements determine the same coset precisely when their difference belongs to N:

$$m + N = m' + N \iff m - m' \in N.$$

Thus N defines an equivalence relation on M:

$$m \sim m' \iff m - m' \in N.$$

The set of equivalence classes will become the quotient module.

1.16 Quotient Modules

Definition 1.16.1. Let M be an R-module and $N \leq M$. The quotient module M/N is the set of cosets

$$M/N = \{m + N : m \in M\}.$$

Addition is defined by

$$(m + N) + (m' + N) = (m + m') + N,$$

and scalar multiplication by

$$r(m+N) = rm + N.$$

These operations are well defined.
Hence

$$M/N$$

is again an R-module.
This directly generalizes quotient vector spaces.

Example 1.16.2. As a $\mathbb{Z}$-module,

$$\mathbb{Z}/n\mathbb{Z}$$

is precisely the quotient module

$$\mathbb{Z}/n\mathbb{Z}.$$

Thus one of the most familiar finite abelian groups is naturally a quotient module.

1.17 The Canonical Projection

Associated with every quotient module is a natural map

$$\pi : M \to M/N$$

defined by

$$\pi(m) = m + N.$$

This map is surjective and satisfies

$$\ker \pi = N.$$

Therefore

$$\boxed{N \hookrightarrow M \twoheadrightarrow M/N}$$

encodes the relationship among a module, a submodule, and its quotient.
This pattern will later be expressed more systematically using exact sequences.

1.18 The First Isomorphism Theorem

The familiar first isomorphism theorem from linear algebra survives in module theory.

Theorem 1.18.1 (First Isomorphism Theorem)**.** Let

$$f : M \to N$$

be an R-module homomorphism. Then

$$\boxed{M/\ker f \cong_R \operatorname{Im} f.}$$

Proof. Define

$$\overline{f} : M/\ker f \to \operatorname{Im} f$$

by

$$\overline{f}(m + \ker f) = f(m).$$

If

$$m + \ker f = m' + \ker f,$$

then

$$m - m' \in \ker f.$$

Hence

$$f(m) = f(m'),$$

so the map is well defined.
It is straightforward to verify that $\overline{f}$ is an R-module homomorphism.
It is surjective by the definition of $\operatorname{Im} f$.
If

$$\overline{f}(m + \ker f) = 0,$$

then

$$f(m) = 0,$$

so

$$m \in \ker f.$$

Therefore

$$m + \ker f = \ker f,$$

which is the zero element of the quotient.
Hence $\overline{f}$ is injective and therefore an isomorphism. □

The structural content is

$$\boxed{\text{domain/kernel} \cong \text{image.}}$$

1.19 The Second Isomorphism Theorem

Theorem 1.19.1 (Second Isomorphism Theorem)**.** Let $A, B \leq M$. Then

$$A + B$$

is a submodule,

$$A \cap B$$

is a submodule of A, and

$$\boxed{A/(A \cap B) \cong_R (A + B)/B.}$$

The theorem compares two ways of measuring the part of A that is not already contained in B.

1.20 The Third Isomorphism Theorem

Theorem 1.20.1 (Third Isomorphism Theorem)**.** Suppose

$$N \leq L \leq M.$$

Then

$$L/N$$

is a submodule of

$$M/N,$$

and

$$\boxed{(M/N)/(L/N) \cong_R M/L.}$$

Thus quotienting in two stages is equivalent to quotienting directly by the larger submodule.
Schematically,

$$M \longrightarrow M/N \longrightarrow (M/N)/(L/N)$$

has the same final quotient structure as

$$M \longrightarrow M/L.$$

1.21 Correspondence of Submodules

Let

$$N \leq M.$$

Submodules of M/N correspond naturally to submodules of M containing N.
More precisely,

$$L \longmapsto L/N$$

gives a bijection

$$\boxed{\{L \leq M : N \subseteq L\} \longleftrightarrow \{\text{submodules of } M/N\}.}$$

This is the module version of the correspondence theorem for quotient spaces and quotient groups.

1.22 Torsion Elements

Module theory contains phenomena that do not appear in vector spaces.

Definition 1.22.1. Let R be an integral domain and M an R-module. An element $m \in M$ is a torsion element if there exists a nonzero

$$r \in R$$

such that

$$rm = 0.$$

The set of torsion elements is denoted by

$$\mathrm{Tor}(M).$$

For modules over an integral domain,

$$\mathrm{Tor}(M)$$

is a submodule.

Example 1.22.2. Every nonzero element of

$$\mathbb{Z}/n\mathbb{Z}$$

is a torsion element as a $\mathbb{Z}$-module.

By contrast,

$$\mathbb{Z}$$

is torsion-free as a $\mathbb{Z}$-module.
A vector space over a field has no nonzero torsion elements because

$$av = 0, \qquad a \neq 0,$$

implies

$$v = a^{-1}(av) = 0.$$

Thus torsion is one of the characteristic phenomena introduced when a field is replaced by a ring.

1.23 Torsion-Free Modules

Definition 1.23.1. An R-module M over an integral domain is torsion-free if

$$rm = 0, \qquad r \neq 0,$$

implies

$$m = 0.$$

Every free module over an integral domain is torsion-free.
However, the converse need not hold over an arbitrary integral domain.
This provides another warning that familiar equivalences from vector spaces do not automatically survive in module theory.

1.24 Modules over a Principal Ideal Domain

A particularly well-behaved class of rings is formed by principal ideal domains.
Examples include

$$\mathbb{Z}$$

and

$$K[x]$$

when K is a field.
Modules over principal ideal domains retain enough structure to admit strong classification results.
For example, finitely generated modules over a principal ideal domain can be decomposed into a free part and a torsion part.
The complete structure theorem belongs to a broader development of module theory. At this stage, the important point is the transition

$$\boxed{\text{vector spaces over fields} \longrightarrow \text{modules over PIDs} \longrightarrow \text{modules over general rings.}}$$

As the scalar ring becomes more general, the theory becomes less rigid and new algebraic phenomena appear.

1.25 Vector Spaces and Modules Compared

The analogy between vector spaces and modules may be summarized as follows:

Vector Spaces	Modules
field K	ring R
vector subspace	submodule
span	generated submodule
linear transformation	module homomorphism
quotient space	quotient module
basis	basis of a free module
dimension	rank when appropriate

The formal similarities are substantial, but the differences are just as important.
For finite-dimensional vector spaces:

$$\boxed{\text{every vector space has a basis,}}$$

$$\boxed{\text{every subspace has a complement,}}$$

and

$$\boxed{\text{dimension controls isomorphism.}}$$

For general modules, none of these principles can be used without additional assumptions.

1.26 Why Modules Matter

Modules provide a common language for several familiar objects.
A vector space is a module over a field.
An abelian group is a module over $\mathbb{Z}$.
An ideal is a submodule of a ring considered as a module over itself.
Many algebraic constructions that appear different at first can therefore be studied through the same language.
The conceptual transition is

$$\boxed{\text{linear algebra over fields} \longrightarrow \text{linear algebra over rings.}}$$

But this phrase should be used with caution: module theory is not merely ordinary linear algebra with the word "field" replaced by "ring". The failure of scalar invertibility changes the structure of the theory fundamentally.

1.27 From Quotients to Exact Sequences

The canonical quotient map

$$\pi : M \to M/N$$

satisfies

$$\ker \pi = N.$$

If

$$\iota : N \to M$$

denotes the inclusion map, then

$$\operatorname{Im} \iota = N = \ker \pi.$$

Thus the relationship among N, M, and M/N can be written as

$$0 \longrightarrow N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \longrightarrow 0.$$

This notation records simultaneously that

$$\iota$$

is injective,

$$\pi$$

is surjective, and

$$\operatorname{Im} \iota = \ker \pi.$$

Such sequences provide a compact language for describing how algebraic objects are assembled from subobjects and quotients.
They form one of the natural next steps beyond the elementary theory of modules developed in this chapter.

Chapter 2

Exact Sequences and Duality

2.1 Sequences of Module Homomorphisms

Let R be a ring, and let

$$M_0, M_1, M_2, \dots$$

be R-modules.
A sequence of module homomorphisms is a diagram

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \longrightarrow \cdots .$$

Such a diagram records not only the modules involved, but also the relationships among them.
The key condition is that everything produced by one map is precisely what is annihilated by the next.

Definition 2.1.1. The sequence

$$M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1}$$

is exact at M_i if

$$\boxed{\operatorname{Im} f_{i-1} = \ker f_i.}$$

A sequence is exact if it is exact at every intermediate module.
Exactness therefore combines the two most basic objects associated with a homomorphism:

$$\text{image} \qquad \text{and} \qquad \text{kernel.}$$

2.2 The Zero Module in an Exact Sequence

The zero module makes injectivity and surjectivity easy to express.
Consider

$$0 \longrightarrow M \xrightarrow{f} N.$$

Since the image of

$$0 \to M$$

is $\{0\}$, exactness at M means

$$\ker f = \{0\}.$$

Therefore

$$\boxed{0 \to M \xrightarrow{f} N \text{ is exact } \iff f \text{ is injective.}}$$

Similarly, consider

$$M \xrightarrow{f} N \longrightarrow 0.$$

The kernel of the zero map

$$N \to 0$$

is all of N. Hence exactness at N means

$$\operatorname{Im} f = N.$$

Thus

$$\boxed{M \xrightarrow{f} N \to 0 \text{ is exact } \iff f \text{ is surjective.}}$$

These are among the basic observations explicitly recorded in the original notes. :contentReference[oaicite:2]index=2

2.3 Kernel and Cokernel Sequences

Let

$$f : M \to N$$

be an R-module homomorphism.
The inclusion

$$\iota : \ker f \hookrightarrow M$$

gives an exact sequence

$$0 \longrightarrow \ker f \xrightarrow{\iota} M \xrightarrow{f} N.$$

Indeed,

$$\operatorname{Im} \iota = \ker f.$$

There is a dual construction on the other side.

Definition 2.3.1. The cokernel of f is

$$\operatorname{coker} f = N/\operatorname{Im} f.$$

Let

$$\pi : N \to \operatorname{coker} f$$

be the canonical quotient map. Then

$$M \xrightarrow{f} N \xrightarrow{\pi} \operatorname{coker} f \longrightarrow 0$$

is exact.
Combining the two constructions gives

$$0 \longrightarrow \ker f \longrightarrow M \xrightarrow{f} N \longrightarrow \operatorname{coker} f \longrightarrow 0,$$

which is exact everywhere except that one must remember that

$$\operatorname{Im} f$$

is generally only a submodule of N.
The cokernel measures the failure of f to be surjective:

$$\operatorname{coker} f = 0 \iff f \text{ is surjective.}$$

Similarly,

$$\ker f = 0 \iff f \text{ is injective.}$$

2.4 Short Exact Sequences

Definition 2.4.1. A short exact sequence is an exact sequence of the form

$$\boxed{0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0.}$$

Exactness means simultaneously that:

$$f \text{ is injective,}$$

$$g \text{ is surjective,}$$

and

$$\operatorname{Im} f = \ker g.$$

Thus A can be identified with a submodule of B, while C is isomorphic to the corresponding quotient.
More precisely,

$$C \cong B/\operatorname{Im} f.$$

Hence a short exact sequence expresses the structural idea

$$\boxed{B \text{ is built from } A \text{ and } C.}$$

The precise meaning of "built from" may nevertheless be nontrivial: B need not be a direct sum of A and C.

2.5 The Standard Quotient Sequence

Let

$$N \leq M.$$

Let

$$\iota : N \to M$$

be the inclusion map and

$$\pi : M \to M/N$$

the canonical projection.
Then

$$\boxed{0 \longrightarrow N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \longrightarrow 0}$$

is exact.
Indeed,

$$\ker \pi = N = \operatorname{Im} \iota.$$

This short exact sequence is the natural algebraic encoding of a submodule and its quotient.
It is also one of the examples explicitly appearing in the original course notes. :contentReference[oaicite:3]index=3

2.6 Exactness and the First Isomorphism Theorem

Suppose

$$M \xrightarrow{f} N$$

is a homomorphism.
The first isomorphism theorem gives

$$M/\ker f \cong \operatorname{Im} f.$$

Thus the sequence

$$0 \longrightarrow \ker f \longrightarrow M \xrightarrow{f} \operatorname{Im} f \longrightarrow 0$$

is short exact.
Exact sequences therefore package the first isomorphism theorem into a diagrammatic form.
Instead of repeatedly writing

$$M/\ker f \cong \operatorname{Im} f,$$

one may regard

$$0 \to \ker f \to M \to \operatorname{Im} f \to 0$$

as the structural object.

2.7 Split Short Exact Sequences

A short exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

need not imply

$$B \cong A \oplus C.$$

When such a decomposition does occur, the sequence is said to split.

Definition 2.7.1. The short exact sequence

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

splits if there exists a homomorphism

$$s : C \to B$$

such that

$$\boxed{g \circ s = \mathrm{id}_C\,.}$$

The map s is called a section of g.
Thus s chooses, in a homomorphic way, one representative in B for every element of C.
There is an equivalent formulation using the injection f.

Definition 2.7.2. A retraction of f is a homomorphism

$$r : B \to A$$

such that

$$\boxed{r \circ f = \mathrm{id}_A\,.}$$

2.8 Splitting Lemma

Theorem 2.8.1 (Splitting Lemma)**.** Let

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

be a short exact sequence.
The following are equivalent:

1. there exists a section

   $$s : C \to B$$

   satisfying

   $$g \circ s = \mathrm{id}_C;$$

2. there exists a retraction
$$r : B \to A$$
satisfying
$$r \circ f = \mathrm{id}_A;$$
3.
$$B \cong A \oplus C.$$

Suppose a section s exists. Then
$$\operatorname{Im} s \cap \operatorname{Im} f = \{0\}.$$
Indeed, if
$$s(c) = f(a),$$
then applying g gives
$$c = g(s(c)) = g(f(a)) = 0.$$
Hence
$$s(c) = 0.$$
Also, for every $b \in B$,
$$b - s(g(b)) \in \ker g = \operatorname{Im} f.$$
Therefore
$$B = \operatorname{Im} f \oplus \operatorname{Im} s.$$
Since
$$\operatorname{Im} f \cong A$$
and
$$\operatorname{Im} s \cong C,$$
we obtain
$$B \cong A \oplus C.$$

2.9 Why Splitting Is Automatic for Vector Spaces

Let
$$0 \longrightarrow U \longrightarrow V \longrightarrow W \longrightarrow 0$$
be a short exact sequence of vector spaces.
Since every subspace of a vector space has a complement, we may write
$$V = U \oplus U'.$$
The quotient map restricts to an isomorphism
$$U' \cong W.$$
Hence
$$\boxed{V \cong U \oplus W.}$$
Therefore every short exact sequence of vector spaces splits.
This is another point where module theory differs sharply from ordinary linear algebra. For general modules, a submodule need not have a complement, and a short exact sequence need not split.

2.10 A Nonsplitting Example

Consider the short exact sequence of $\mathbb{Z}$-modules

$$0 \longrightarrow 2\mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

If this sequence split, then

$$\mathbb{Z} \cong 2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}.$$

But the right-hand side contains a nonzero torsion element, while $\mathbb{Z}$ is torsion-free.
Therefore the sequence does not split.
Thus exactness alone does not imply direct-sum decomposition.

2.11 Exact Sequences as Extension Data

A short exact sequence

$$0 \to A \to B \to C \to 0$$

may be viewed as describing an extension of C by A.
The module B contains a copy of A, and after collapsing that copy one obtains C:

$$B/A \cong C.$$

If the sequence splits, then the extension is structurally simple:

$$B \cong A \oplus C.$$

If it does not split, B contains additional information describing how A and C have been assembled.
Thus

$$\boxed{\text{short exact sequence} = \text{subobject} + \text{quotient} + \text{extension data.}}$$

This perspective becomes increasingly important in later algebra.

2.12 Dual Modules

We now return temporarily to modules over a field K, where duality behaves particularly cleanly.
Let V be a K-vector space.

Definition 2.12.1. The dual space of V is

$$V^* = \mathrm{Hom}_K(V, K).$$

Thus elements of V^* are linear functionals

$$\varphi : V \to K.$$

For general R-modules, one may similarly define

$$M^* = \mathrm{Hom}_R(M, R),$$

subject to the appropriate left/right module conventions.
The original notes introduce the dual module precisely through

$$M^* = \mathrm{Hom}(M, K)$$

in the vector-space setting. :contentReference[oaicite:4]index=4

2.13 The Dual Map

Let

$$f : V \to W$$

be linear.

Definition 2.13.1. The dual map of f is

$$f^* : W^* \to V^*$$

defined by

$$\boxed{f^*(\varphi) = \varphi \circ f.}$$

Thus duality reverses the direction of a map:

$$V \xrightarrow{f} W$$

becomes

$$W^* \xrightarrow{f^*} V^*.$$

This reversal is the defining feature of duality.

2.14 Contravariance of the Dual

Dual maps satisfy

$$(\mathrm{id}_V)^* = \mathrm{id}_{V^*}.$$

If

$$U \xrightarrow{f} V \xrightarrow{g} W,$$

then

$$(g \circ f)^* = f^* \circ g^*.$$

Thus the order of composition is reversed:

$$\boxed{(g \circ f)^* = f^* \circ g^*.}$$

The original notes explicitly record this contravariant behavior of the dual operation. :contentReference[oaicite:5]index=5

2.15 Dual Bases

Let

$$\mathcal{B} = (v_1, \dots, v_n)$$

be a basis of a finite-dimensional vector space V.

Definition 2.15.1. The dual basis

$$\mathcal{B}^* = (v_1^*, \dots, v_n^*)$$

of V^* is defined by

$$\boxed{v_i^*(v_j) = \delta_{ij}.}$$

If

$$v = a_1 v_1 + \cdots + a_n v_n,$$

then

$$v_i^*(v) = a_i.$$

Thus the dual basis extracts coordinates.
Every functional

$$\varphi \in V^*$$

has the representation

$$\varphi = \sum_{i=1}^{n} \varphi(v_i) v_i^*.$$

Therefore

$$\dim V^* = \dim V.$$

2.16 Matrix of the Dual Map

Let

$$f : V \to W$$

be linear.
Choose bases

$$\mathcal{B} = (v_1, \ldots, v_n)$$

of V and

$$\mathcal{C} = (w_1, \ldots, w_m)$$

of W.
Suppose

$$[f]_{\mathcal{C} \leftarrow \mathcal{B}} = A.$$

With respect to the corresponding dual bases,

$$\boxed{[f^*]_{\mathcal{B}^* \leftarrow \mathcal{C}^*} = A^T.}$$

Thus transposition is the matrix manifestation of the dual map.
This should be distinguished from the adjoint studied in *Advanced Linear Algebra II*. The dual map requires no inner product, while the adjoint does.

2.17 Duality and Kernels

Let

$$f : V \to W.$$

A functional

$$\varphi \in W^*$$

belongs to $\ker f^*$ precisely when

$$\varphi \circ f = 0.$$

Equivalently,

$$\varphi$$

vanishes on

$$\operatorname{Im} f.$$

This motivates the annihilator.

Definition 2.17.1. For a subspace $U \subseteq V$, the annihilator of U is

$$U^\circ = \{\varphi \in V^* : \varphi(u) = 0 \text{ for every } u \in U\}.$$

Then

$$\boxed{\ker f^* = (\operatorname{Im} f)^\circ.}$$

Similarly,

$$\operatorname{Im} f^* = (\ker f)^\circ$$

in finite dimensions.

2.18 Dimension of the Annihilator

Let V be finite-dimensional and

$$U \subseteq V.$$

Then

$$\boxed{\dim U^\circ = \dim V - \dim U.}$$

To see this, choose a basis

$$u_1, \dots, u_r$$

of U and extend it to a basis

$$u_1, \dots, u_r, v_{r+1}, \dots, v_n$$

of V.
In the dual basis,

$$U^\circ = \operatorname{span}\{v_{r+1}^*, \dots, v_n^*\}.$$

Hence

$$\dim U^\circ = n - r.$$

This is the dual-space analogue of the codimension of a subspace.

2.19 Double Duals

The dual space may itself be dualized.
For $v \in V$, define

$$\widehat{v} : V^* \to K$$

by

$$\widehat{v}(\varphi) = \varphi(v).$$

This defines a linear map

$$\iota_V : V \to V^{**}, \qquad v \mapsto \widehat{v}.$$

Theorem 2.19.1. The canonical map

$$\iota_V : V \to V^{**}$$

is injective.

If V is finite-dimensional, then

$$\dim V = \dim V^{**}.$$

Therefore:

Corollary 2.19.2. For finite-dimensional V,

$$\boxed{V \cong V^{**}}$$

canonically.

The word "canonically" matters: the isomorphism

$$V \to V^{**}$$

does not require a choice of basis.
By contrast, an isomorphism

$$V \cong V^*$$

usually requires additional choices or structure.

2.20 Dual of a Direct Sum

For a finite collection of vector spaces,

$$V_1, \dots, V_r,$$

we have

$$\boxed{(V_1 \oplus \cdots \oplus V_r)^* \cong V_1^* \oplus \cdots \oplus V_r^*.}$$

A functional on the direct sum is determined uniquely by its restrictions to each summand. For infinite families, the relationship between direct sums and direct products becomes subtler. The original notes explicitly flag this distinction while discussing dual spaces. :contentReference[oaicite:6]index=6
We do not develop the infinite-dimensional case systematically here.

2.21 Dualizing a Short Exact Sequence

Suppose

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is a short exact sequence of finite-dimensional vector spaces.
Dualizing reverses the arrows:

$$0 \longleftarrow A^* \xleftarrow{f^*} B^* \xleftarrow{g^*} C^* \longleftarrow 0.$$

Writing the arrows in the usual left-to-right direction gives

$$\boxed{0 \longrightarrow C^* \xrightarrow{g^*} B^* \xrightarrow{f^*} A^* \longrightarrow 0.}$$

This sequence is again exact.
The original course notes explicitly record this reversal of a short exact sequence under dualization in the vector-space setting. :contentReference[oaicite:7]index=7
Thus duality converts

$$A \hookrightarrow B \twoheadrightarrow C$$

into

$$C^* \hookrightarrow B^* \twoheadrightarrow A^*.$$

2.22 Why Duality Reverses Exact Sequences

Consider

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0.$$

Since g is surjective, if

$$g^*(\varphi) = 0,$$

then

$$\varphi \circ g = 0.$$

Because every element of C lies in the image of g,

$$\varphi = 0.$$

Hence g^* is injective.
Similarly, since f is injective in the finite-dimensional vector-space setting, every functional on A can be extended to a functional on B. Therefore

$$f^* : B^* \to A^*$$

is surjective.
Finally,

$$\operatorname{Im} g^* = \ker f^*.$$

Thus exactness survives, but the direction is reversed.

2.23 Exactness and Dimension

For finite-dimensional vector spaces, a short exact sequence

$$0 \to A \to B \to C \to 0$$

implies

$$\boxed{\dim B = \dim A + \dim C.}$$

Indeed,

$$C \cong B/A,$$

so

$$\dim C = \dim B - \dim A.$$

Thus the familiar dimension formula is the numerical shadow of a short exact sequence. For general modules, there is usually no ordinary dimension function, so exact sequences become more fundamental than dimension counting.

2.24 Exact Sequences as Structural Language

The point of exact sequences is not merely notation.
The expression

$$0 \to A \to B \to C \to 0$$

simultaneously records:

$$A \text{ sits inside } B,$$

$$C \text{ is the corresponding quotient,}$$

and

$$B/A \cong C.$$

If the sequence splits, then

$$B \cong A \oplus C.$$

If it does not split, the middle term contains additional extension information.
Duality then reverses this structure:

$$\boxed{\text{subobjects} \longleftrightarrow \text{quotient information.}}$$

This makes exact sequences one of the most efficient organizational languages in modern algebra.

2.25 From Exact Sequences to Tensor Products

The present chapter has introduced two structural ideas.
First, exact sequences encode how modules are related through kernels, images, submodules, and quotients:

$$\boxed{\text{kernel} \longrightarrow \text{module} \longrightarrow \text{quotient.}}$$

Second, duality converts maps into maps in the opposite direction:

$$\boxed{V \xrightarrow{f} W \quad \longrightarrow \quad W^* \xrightarrow{f^*} V^*.}$$

The next major construction begins from a different problem.
Suppose

$$B : M \times N \to P$$

is bilinear.
Rather than studying bilinear maps directly, one would like to build a module $M \otimes_R N$ for which bilinear maps from $M \times N$ correspond to ordinary linear maps from $M \otimes_R N$. This universal construction is the tensor product and forms the subject of the next chapter.

Chapter 3

Tensor Products and Multilinear Algebra

3.1 Why Tensor Products?

Let K be a field, and let V, W, X be vector spaces over K.
A linear map

$$T : V \to X$$

is controlled by one variable.
A bilinear map

$$B : V \times W \to X$$

is linear in each variable separately:

$$B(av_1 + bv_2, w) = aB(v_1, w) + bB(v_2, w),$$

and

$$B(v, aw_1 + bw_2) = aB(v, w_1) + bB(v, w_2).$$

Bilinear maps occur naturally throughout algebra. Examples include matrix multiplication, bilinear forms, evaluation pairings, and multiplication in an algebra.
The tensor product is designed to replace a bilinear problem by a linear one.
The guiding principle is

$$\boxed{\text{bilinear maps } V \times W \to X \longleftrightarrow \text{linear maps } V \otimes W \to X.}$$

Thus $V \otimes W$ is not introduced merely as another vector space. Its defining purpose is to linearize bilinear maps.

3.2 Formal Symbols

To construct $V \otimes W$, begin with formal symbols

$$v \otimes w, \qquad v \in V, \quad w \in W.$$

At this stage,

$$v \otimes w$$

is only a symbol. It should not be interpreted as ordinary multiplication.
We want these symbols to satisfy the bilinear relations

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w,$$

$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2,$$

and

$$(av) \otimes w = a(v \otimes w) = v \otimes (aw).$$

These relations encode precisely the bilinearity we want the map

$$(v, w) \mapsto v \otimes w$$

to possess.

3.3 Construction of the Tensor Product

Let F be the free vector space generated by the set

$$V \times W.$$

Thus the elements of F are finite formal linear combinations

$$\sum_{i=1}^{r} a_i[v_i, w_i].$$

Let $R \subseteq F$ be the subspace generated by all elements of the following forms:

$$[v_1 + v_2, w] - [v_1, w] - [v_2, w],$$

$$[v, w_1 + w_2] - [v, w_1] - [v, w_2],$$

$$[av, w] - a[v, w],$$

and

$$[v, aw] - a[v, w].$$

Definition 3.3.1. The tensor product of V and W is the quotient space

$$\boxed{V \otimes_K W = F/R.}$$

The equivalence class of $[v, w]$ is denoted by

$$v \otimes w.$$

The quotient construction forces all bilinear relations to hold.
This is essentially the construction recorded in the original notes: start with formal symbols and quotient by the relations expressing linearity in each variable. :contentReference[oaicite:3]index=3

3.4 Basic Tensor Identities

By construction,

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w,$$

$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2,$$

and

$$(av) \otimes w = a(v \otimes w) = v \otimes (aw).$$

In particular,

$$0 \otimes w = 0$$

and

$$v \otimes 0 = 0.$$

Also,

$$(-v) \otimes w = -(v \otimes w).$$

Every element of $V \otimes W$ is a finite sum

$$\sum_{i=1}^{r} v_i \otimes w_i.$$

However, not every tensor can be written as a single pure tensor

$$v \otimes w.$$

This distinction is important.

3.5 Pure Tensors

Definition 3.5.1. An element of $V \otimes W$ of the form

$$v \otimes w$$

is called a pure tensor or decomposable tensor.

A general tensor is a finite sum of pure tensors.
For example,

$$v_1 \otimes w_1 + v_2 \otimes w_2$$

need not equal

$$v \otimes w$$

for any v, w.
Thus

$$V \otimes W$$

is generated by pure tensors, but it consists of more than pure tensors alone.

3.6 The Canonical Bilinear Map

Define

$$\tau : V \times W \to V \otimes W$$

by

$$\boxed{\tau(v, w) = v \otimes w.}$$

By construction, τ is bilinear.
The tensor product is characterized by the fact that every bilinear map factors uniquely through τ.
This is the universal property of the tensor product.

3.7 Universal Property

Theorem 3.7.1 (Universal Property of the Tensor Product)**.** Let

$$B : V \times W \to X$$

be a bilinear map.
Then there exists a unique linear map

$$\widetilde{B} : V \otimes W \to X$$

such that

$$\boxed{B = \widetilde{B} \circ \tau.}$$

Equivalently,

$$\boxed{\widetilde{B}(v \otimes w) = B(v, w).}$$

The corresponding diagram is

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & V \otimes W \\ & \searrow B & \downarrow \widetilde{B} \\ & & X. \end{array}$$

The original notes state exactly this universal property: a bilinear map from $M \times N$ factors uniquely through a linear map from $M \otimes N$. :contentReference[oaicite:4]index=4

3.8 Proof of the Universal Property

Let

$$B : V \times W \to X$$

be bilinear.
First define a linear map

$$F \to X$$

on the free vector space by

$$[v, w] \mapsto B(v, w).$$

Because B is bilinear, every defining relation of R is sent to zero.
For example,

$$B(v_1 + v_2, w) - B(v_1, w) - B(v_2, w) = 0.$$

Similarly,

$$B(v, w_1 + w_2) - B(v, w_1) - B(v, w_2) = 0.$$

Therefore

$$R \subseteq \ker F.$$

Hence F descends to a well-defined linear map on the quotient

$$V \otimes W = F/R.$$

Define

$$\widetilde{B}(v \otimes w) = B(v, w).$$

Since pure tensors span $V \otimes W$, this condition determines $\widetilde{B}$ uniquely.
Thus existence and uniqueness both follow.

3.9 Uniqueness of the Tensor Product

The universal property determines the tensor product uniquely up to a unique isomorphism.
Suppose

$$(T, \tau)$$

and

$$(T', \tau')$$

both satisfy the universal property.
Then there exists a unique linear map

$$f : T \to T'$$

such that

$$f \circ \tau = \tau'.$$

Similarly, there exists a unique linear map

$$g : T' \to T$$

such that

$$g \circ \tau' = \tau.$$

Then

$$g \circ f : T \to T$$

satisfies

$$(g \circ f) \circ \tau = \tau.$$

By uniqueness,

$$g \circ f = \mathrm{id}_T \,.$$

Similarly,

$$f \circ g = \mathrm{id}_{T'} \,.$$

Therefore

$$\boxed{T \cong T'.}$$

Thus the tensor product is unique up to unique isomorphism.
This point is also explicitly recorded in the original notes. :contentReference[oaicite:5]index=5

3.10 Tensor Products of Bases

Let

$$\mathcal{B} = (v_1, \ldots, v_m)$$

be a basis of V, and let

$$\mathcal{C} = (w_1, \ldots, w_n)$$

be a basis of W.

Theorem 3.10.1. The collection

$$\boxed{\{v_i \otimes w_j : 1 \le i \le m,\ 1 \le j \le n\}}$$

is a basis of $V \otimes W$.

The original notes explicitly state this basis theorem in the form that bases B and C produce a basis

$$\{b_i \otimes c_j\}$$

of the tensor product. :contentReference[oaicite:6]index=6

Proof. Every $v \in V$ and $w \in W$ can be written as

$$v = \sum_{i=1}^{m} a_i v_i, \qquad w = \sum_{j=1}^{n} b_j w_j.$$

Hence

$$v \otimes w = \left(\sum_{i=1}^{m} a_i v_i \right) \otimes \left(\sum_{j=1}^{n} b_j w_j \right).$$

By bilinearity,

$$v \otimes w = \sum_{i=1}^{m} \sum_{j=1}^{n} a_i b_j (v_i \otimes w_j).$$

Therefore the stated tensors span $V \otimes W$.
To prove linear independence, for each pair (i, j) define a bilinear map

$$B_{ij} : V \times W \to K$$

by

$$B_{ij}(v_p, w_q) = \delta_{ip}\delta_{jq}$$

and extend bilinearly.
The universal property gives a linear functional

$$\widetilde{B}_{ij} : V \otimes W \to K.$$

If

$$\sum_{p,q} c_{pq} v_p \otimes w_q = 0,$$

applying $\widetilde{B}_{ij}$ gives

$$c_{ij} = 0.$$

Hence all coefficients vanish. □

3.11 Dimension Formula

If

$$\dim V = m$$

and

$$\dim W = n,$$

then the basis theorem gives

$$\boxed{\dim(V \otimes W) = mn.}$$

Thus tensor product multiplies dimensions:

$$\boxed{\dim(V \otimes W) = (\dim V)(\dim W).}$$

For example,

$$K^m \otimes K^n \cong K^{mn}.$$

The isomorphism is not canonical until bases or another natural identification are specified.

3.12 Simple Examples

Since K is one-dimensional over itself,

$$K \otimes V \cong V.$$

An explicit isomorphism is

$$K \otimes V \to V, \qquad a \otimes v \mapsto av.$$

Similarly,

$$V \otimes K \cong V.$$

Also,

$$K^m \otimes K^n$$

has basis

$$e_i \otimes f_j,$$

so

$$K^m \otimes K^n \cong K^{mn}.$$

3.13 Tensor Products and Matrices

Let

$$V = K^m, \qquad W = K^n.$$

A tensor

$$T \in V \otimes W$$

can be written uniquely as

$$T = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} e_i \otimes f_j.$$

Thus T can be represented by the matrix

$$A = (a_{ij}).$$

This gives, after choosing the standard bases, a vector-space isomorphism

$$\boxed{K^m \otimes K^n \cong M_{m \times n}(K).}$$

Under this identification, a pure tensor

$$v \otimes w$$

corresponds to a rank-one matrix

$$vw^T.$$

Thus the distinction between pure and general tensors parallels the distinction between rank-one matrices and arbitrary matrices.

3.14 Tensor Products and Dual Spaces

For finite-dimensional vector spaces, tensor products are closely related to spaces of linear maps.

Let V and W be finite-dimensional.

Given

$$w \in W$$

and

$$\varphi \in V^*,$$

define

$$T_{w,\varphi} : V \to W$$

by

$$T_{w,\varphi}(v) = \varphi(v) w.$$

This map has rank at most one.

The assignment

$$(w, \varphi) \mapsto T_{w,\varphi}$$

is bilinear.

Hence, by the universal property, it induces a linear map

$$W \otimes V^* \to \mathrm{Hom}_K(V, W).$$

In finite dimensions, this map is an isomorphism:

$$\boxed{W \otimes V^* \cong \mathrm{Hom}_K(V, W).}$$

This is one of the structural identifications reflected in the tensor/Hom summary of the original notes. :contentReference[oaicite:7]index=7

3.15 Endomorphisms as Tensors

Taking

$$W = V,$$

we obtain

$$\boxed{V \otimes V^* \cong \mathrm{End}_K(V).}$$

A pure tensor

$$v \otimes \varphi$$

corresponds to the rank-one operator

$$x \mapsto \varphi(x)v.$$

Thus every finite-dimensional linear operator can be written as a sum of rank-one operators.
If

$$(e_1, \ldots, e_n)$$

is a basis and

$$(e_1^*, \ldots, e_n^*)$$

its dual basis, then

$$e_i \otimes e_j^*$$

corresponds to the elementary matrix E_{ij}.
Hence the usual matrix basis is naturally a tensor-product basis.

3.16 Tensor Product of Linear Maps

Let

$$f : V \to V'$$

and

$$g : W \to W'$$

be linear maps.
The map

$$V \times W \to V' \otimes W'$$

defined by

$$(v, w) \mapsto f(v) \otimes g(w)$$

is bilinear.
Therefore it induces a unique linear map

$$\boxed{f \otimes g : V \otimes W \to V' \otimes W'}$$

satisfying

$$(f \otimes g)(v \otimes w) = f(v) \otimes g(w).$$

Tensor products therefore act not only on vector spaces but also on linear maps.

3.17 Functorial Properties

Tensor products of maps satisfy

$$\mathrm{id}_V \otimes \mathrm{id}_W = \mathrm{id}_{V\otimes W}\,.$$

If

$$f_1 : V \to V', \qquad f_2 : V' \to V'',$$

and

$$g_1 : W \to W', \qquad g_2 : W' \to W'',$$

then

$$\boxed{(f_2 \circ f_1) \otimes (g_2 \circ g_1) = (f_2 \otimes g_2) \circ (f_1 \otimes g_1).}$$

Also,

$$(f_1 + f_2) \otimes g = f_1 \otimes g + f_2 \otimes g,$$

whenever the expressions are defined.
These identities follow by checking pure tensors, which span the tensor product.

3.18 Tensor Products and Direct Sums

Tensor products distribute over finite direct sums.
For example,

$$\boxed{(V_1 \oplus V_2) \otimes W \cong (V_1 \otimes W) \oplus (V_2 \otimes W).}$$

Similarly,

$$\boxed{V \otimes (W_1 \oplus W_2) \cong (V \otimes W_1) \oplus (V \otimes W_2).}$$

The first isomorphism is determined on pure tensors by

$$(v_1, v_2) \otimes w \mapsto (v_1 \otimes w,\, v_2 \otimes w).$$

Thus tensor products interact naturally with direct-sum decompositions.

3.19 Multiple Tensor Products

The tensor construction extends to more than two vector spaces.
Let

$$V_1, \ldots, V_d$$

be vector spaces.
Their tensor product

$$V_1 \otimes \cdots \otimes V_d$$

is characterized by a multilinear map

$$\tau : V_1 \times \cdots \times V_d \to V_1 \otimes \cdots \otimes V_d$$

such that every multilinear map

$$F : V_1 \times \cdots \times V_d \to X$$

factors uniquely through a linear map

$$\widetilde{F} : V_1 \otimes \cdots \otimes V_d \to X.$$

Thus

$$\boxed{F = \widetilde{F} \circ \tau.}$$

The original notes explicitly extend the universal construction from bilinear maps to multilinear maps of several variables. :contentReference[oaicite:8]index=8

3.20 Associativity of Tensor Products

There are natural isomorphisms

$$(V \otimes W) \otimes X \cong V \otimes (W \otimes X).$$

On pure tensors,

$$(v \otimes w) \otimes x \longmapsto v \otimes (w \otimes x).$$

Because of this canonical identification, parentheses are usually suppressed and we write

$$V \otimes W \otimes X.$$

More generally,

$$V_1 \otimes \cdots \otimes V_d$$

can be written without specifying an order of parenthesization.

3.21 Tensor Powers

Definition 3.21.1. For a vector space V, the d-th tensor power is

$$\boxed{V^{\otimes d} = \underbrace{V \otimes \cdots \otimes V}_{d \text{ factors}}.}$$

For $d = 0$, one conventionally sets

$$V^{\otimes 0} = K.$$

If

$$\dim V = n,$$

then

$$\dim V^{\otimes d} = n^d.$$

If

$$(e_1, \ldots, e_n)$$

is a basis of V, then

$$e_{i_1} \otimes \cdots \otimes e_{i_d}$$

with

$$1 \leq i_j \leq n$$

forms a basis of $V^{\otimes d}$.
Tensor powers are explicitly recorded in the original notes as part of the multilinear construction. :contentReference[oaicite:9]index=9

3.22 Multilinear Maps

Definition 3.22.1. A map

$$F : V_1 \times \cdots \times V_d \to W$$

is multilinear if it is linear in every variable separately.

For example, for each fixed choice of all variables except the i-th one,

$$v_i \mapsto F(v_1, \dots, v_i, \dots, v_d)$$

must be linear.
By the universal property,

$$\boxed{\mathrm{Mult}(V_1, \dots, V_d; W) \cong \mathrm{Hom}(V_1 \otimes \cdots \otimes V_d, W).}$$

Thus tensor products convert multilinear algebra into ordinary linear algebra.
This is the central conceptual purpose of the construction.

3.23 Multilinear Forms

A multilinear form is a multilinear map

$$F : V_1 \times \cdots \times V_d \to K.$$

When all spaces equal V, we have

$$F : V^d \to K.$$

Such forms correspond to elements of

$$(V^{\otimes d})^*.$$

In finite dimensions,

$$(V^{\otimes d})^* \cong (V^*)^{\otimes d}.$$

Thus a multilinear form may be regarded as a covariant tensor.
This provides one algebraic origin of the terminology "tensor."

3.24 Symmetric Multilinear Maps

Let

$$F : V^d \to W$$

be multilinear.

Definition 3.24.1. The map F is symmetric if

$$F(v_{\sigma(1)}, \dots, v_{\sigma(d)}) = F(v_1, \dots, v_d)$$

for every permutation

$$\sigma \in S_d.$$

Thus permuting the arguments does not change the value.
For $d = 2$, this reduces to the familiar condition

$$F(v, w) = F(w, v).$$

The original notes explicitly distinguish symmetric multilinear maps through invariance under permutations. :contentReference[oaicite:10]index=10

3.25 Antisymmetric and Alternating Maps

Definition 3.25.1. A multilinear map

$$F : V^d \to W$$

is antisymmetric if

$$F(v_{\sigma(1)}, \dots, v_{\sigma(d)}) = \operatorname{sgn}(\sigma) F(v_1, \dots, v_d)$$

for every

$$\sigma \in S_d.$$

Definition 3.25.2. A multilinear map F is alternating if

$$F(v_1, \dots, v_d) = 0$$

whenever

$$v_i = v_j$$

for some $i \neq j$.

Over a field of characteristic not equal to 2, alternating multilinear maps are antisymmetric.
Indeed, exchanging two arguments changes the sign.
These notions generalize symmetric and alternating bilinear forms.
The original notes record these three classes separately: symmetric, antisymmetric, and alternating multilinear maps. :contentReference[oaicite:11]index=11

3.26 Permutation Action on Tensor Powers

Every permutation

$$\sigma \in S_d$$

acts on

$$V^{\otimes d}$$

by permuting tensor factors:

$$P_\sigma(v_1 \otimes \cdots \otimes v_d) = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(d)}.$$

Thus the symmetric group S_d acts naturally on tensor powers.
This action allows symmetric and alternating tensors to be described algebraically.
The sign

$$\operatorname{sgn}(\sigma) \in \{1, -1\}$$

records whether the permutation is even or odd.

3.27 Symmetric Powers

Definition 3.27.1. The d-th symmetric power of V, denoted by

$$S^d(V),$$

is obtained from $V^{\otimes d}$ by imposing the relations

$$v_1 \otimes \cdots \otimes v_d = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(d)}$$

for every permutation

$$\sigma \in S_d.$$

Equivalently, symmetric tensors do not distinguish the order of their factors.
We denote the symmetric product by

$$v_1 \cdots v_d$$

or, when useful,

$$v_1 \odot \cdots \odot v_d.$$

The original notes define the symmetric power precisely as a quotient of the tensor power by permutation relations. :contentReference[oaicite:12]index=12

3.28 Basis of a Symmetric Power

Let

$$(e_1, \dots, e_n)$$

be a basis of V.
A basis of $S^d(V)$ is given by monomials

$$e_1^{a_1} \cdots e_n^{a_n}$$

with

$$a_1 + \cdots + a_n = d$$

and

$$a_i \geq 0.$$

Therefore

$$\boxed{\dim S^d(V) = \binom{n+d-1}{d}.}$$

For example, if

$$V = \operatorname{span}(x, y, z),$$

then $S^2(V)$ has basis

$$x^2, \quad xy, \quad xz, \quad y^2, \quad yz, \quad z^2.$$

Thus

$$\dim S^2(V) = 6.$$

This corresponds to the 3-dimensional example appearing in the original notes. :contentReference[oaicite:13]index=13

3.29 Exterior Powers

Definition 3.29.1. The d-th exterior power of V, denoted by

$$\bigwedge^d V,$$

is obtained from $V^{\otimes d}$ by imposing the relations

$$v_1 \otimes \cdots \otimes v_d = 0$$

whenever two factors are equal.

The image of

$$v_1 \otimes \cdots \otimes v_d$$

is denoted by

$$v_1 \wedge \cdots \wedge v_d.$$

Thus

$$v \wedge v = 0.$$

Over a field of characteristic not equal to 2,

$$v \wedge w = -w \wedge v.$$

More generally,

$$v_{\sigma(1)} \wedge \cdots \wedge v_{\sigma(d)} = \operatorname{sgn}(\sigma) v_1 \wedge \cdots \wedge v_d.$$

The original notes define exterior powers as quotients of tensor powers by the relations corresponding to repeated factors. :contentReference[oaicite:14]index=14

3.30 Basis of an Exterior Power

Let

$$(e_1, \ldots, e_n)$$

be a basis of V.
Then a basis of

$$\bigwedge^d V$$

is

$$\boxed{e_{i_1} \wedge \cdots \wedge e_{i_d}, \qquad 1 \le i_1 < \cdots < i_d \le n.}$$

Therefore

$$\boxed{\dim \bigwedge^d V = \binom{n}{d}.}$$

In particular,

$$\bigwedge^d V = 0$$

if

$$d > \dim V.$$

For V with basis x, y, z,

$$\bigwedge^2 V$$

has basis

$$x \wedge y, \qquad x \wedge z, \qquad y \wedge z.$$

Hence

$$\dim \bigwedge^2 V = 3.$$

This is the alternating analogue of the symmetric-square example in the notes. :contentReference[oaicite:15]index=15

3.31 Universal Property of Exterior Powers

Exterior powers linearize alternating multilinear maps.
Let

$$F : V^d \to W$$

be alternating and multilinear.
Then there exists a unique linear map

$$\widetilde{F} : \bigwedge^d V \to W$$

such that

$$\boxed{F(v_1, \dots, v_d) = \widetilde{F}(v_1 \wedge \cdots \wedge v_d).}$$

Thus

$$\boxed{\text{alternating multilinear maps on } V^d \longleftrightarrow \text{linear maps from } \bigwedge^d V.}$$

This is the exterior-power version of the universal property of the ordinary tensor product. The notes indicate this same factorization principle after defining symmetric and exterior powers. :contentReference[oaicite:16]index=16

3.32 The Top Exterior Power

Let

$$\dim V = n.$$

Then

$$\dim \bigwedge^n V = 1.$$

If

$$(e_1, \dots, e_n)$$

is a basis of V, then

$$e_1 \wedge \cdots \wedge e_n$$

is a basis of

$$\bigwedge^n V.$$

Let

$$T : V \to V$$

be linear.
Then T induces

$$\bigwedge^n T : \bigwedge^n V \to \bigwedge^n V.$$

Since this space is one-dimensional, the induced map is multiplication by a scalar.
That scalar is

$$\det T.$$

Hence

$$\boxed{\bigwedge^n T = (\det T)\,\mathrm{id}\,.}$$

This gives a conceptual interpretation of the determinant: it is the scalar by which a linear operator acts on the top exterior power.

3.33 Tensor Algebra

The tensor powers of V may be collected into one graded object:

$$T(V) = K \oplus V \oplus V^{\otimes 2} \oplus V^{\otimes 3} \oplus \cdots .$$

Tensor multiplication is defined by concatenation:

$$(v_1 \otimes \cdots \otimes v_r)(w_1 \otimes \cdots \otimes w_s)$$

equals

$$v_1 \otimes \cdots \otimes v_r \otimes w_1 \otimes \cdots \otimes w_s.$$

Thus

$$V^{\otimes r} \cdot V^{\otimes s} \subseteq V^{\otimes (r+s)}.$$

This construction is called the tensor algebra of V.
The symmetric and exterior algebras may be obtained from the tensor algebra by imposing additional relations.

3.34 Why Tensor Products Are Difficult

Tensor products are often difficult on first encounter because the notation

$$v \otimes w$$

looks concrete while its definition is universal and indirect.
Several points should therefore be kept separate.
First,

$$v \otimes w$$

is not ordinary multiplication.
Second, $V \otimes W$ is not the Cartesian product

$$V \times W.$$

Their dimensions already differ:

$$\dim(V \times W) = \dim V + \dim W,$$

while

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

Third, the essential property of $V \otimes W$ is not its explicit quotient construction. Its essential property is the universal factorization

$$\boxed{\text{bilinear} \longrightarrow \text{tensor product} \longrightarrow \text{linear.}}$$

The quotient construction is one method of producing an object with that property.

3.35 Structural Summary

The main chain of this chapter is

$$\boxed{\text{bilinear maps} \longrightarrow V \otimes W \longrightarrow \text{ordinary linear maps.}}$$

For finite-dimensional spaces,

$$\dim(V \otimes W) = \dim V \dim W,$$

and bases combine according to

$$\{v_i\},\ \{w_j\} \quad \longrightarrow \quad \{v_i \otimes w_j\}.$$

Tensor products also produce natural identifications such as

$$W \otimes V^* \cong \mathrm{Hom}(V, W).$$

Repeated tensor products lead to

$$V^{\otimes d},$$

which provides the linear setting for multilinear maps.
Permutation symmetries then lead to two important quotient constructions:

$$\boxed{S^d(V) \quad \text{for symmetric tensors,}}$$

and

$$\boxed{\bigwedge^d V \quad \text{for alternating tensors.}}$$

Thus the conceptual progression is

$$\boxed{\text{linear algebra} \longrightarrow \text{bilinear algebra} \longrightarrow \text{tensor products} \longrightarrow \text{multilinear algebra.}}$$

The material in this chapter is primarily structural. Its purpose in these notes is to preserve the universal constructions and notation needed when tensor products reappear in later algebra, geometry, and related subjects.

Chapter 4

Direct Products, Semidirect Products, and Extensions

4.1 Combining Algebraic Structures

A recurring problem in algebra is to construct a new object from simpler objects.
The simplest examples are direct sums and direct products. If two objects are independent of one another, their operations may be performed componentwise.
More complicated constructions arise when one object acts on another. The resulting structure is no longer a direct product, but may instead be a semidirect product.
The general progression is

independent components $\longrightarrow$ direct product $\longrightarrow$ interaction by an action $\longrightarrow$ semidirect product.

This chapter develops these constructions primarily for groups and then relates them to short exact sequences.

4.2 Direct Products of Groups

Let G and H be groups.

Definition 4.2.1. The direct product of G and H is the set

$$G \times H = \{(g, h) : g \in G,\ h \in H\}$$

with multiplication

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2).$$

The identity element is

$$(e_G, e_H),$$

and

$$(g, h)^{-1} = (g^{-1}, h^{-1}).$$

The two factors occur naturally as subgroups:

$$G \times \{e_H\}$$

and

$$\{e_G\} \times H.$$

Both are normal subgroups of $G \times H$.
Moreover,

$$(G \times \{e_H\}) \cap (\{e_G\} \times H) = \{(e_G, e_H)\}.$$

Every element can be written uniquely as

$$(g, h) = (g, e_H)(e_G, h).$$

Thus a direct product is built from two independent normal factors.

4.3 Projection Maps

The direct product comes with canonical projections

$$\pi_G : G \times H \to G, \qquad \pi_G(g, h) = g,$$

and

$$\pi_H : G \times H \to H, \qquad \pi_H(g, h) = h.$$

Both are group homomorphisms.
Their kernels are

$$\ker \pi_G = \{e_G\} \times H$$

and

$$\ker \pi_H = G \times \{e_H\}.$$

Hence

$$(G \times H)/(\{e_G\} \times H) \cong G,$$

and

$$(G \times H)/(G \times \{e_H\}) \cong H.$$

Thus the direct product may already be expressed through short exact sequences.
For example,

$$1 \longrightarrow G \longrightarrow G \times H \longrightarrow H \longrightarrow 1.$$

4.4 Internal Direct Products

Direct products can also be recognized inside a group.

Definition 4.4.1. Let K be a group and let $G, H \leq K$. We say that K is the internal direct product of G and H if

$$G \trianglelefteq K, \qquad H \trianglelefteq K,$$

$$G \cap H = \{e\},$$

and

$$GH = K.$$

Under these conditions,

$$gh = hg$$

for every $g \in G$ and $h \in H$.
Indeed,

$$ghg^{-1}h^{-1}$$

belongs both to G and to H, so it belongs to

$$G \cap H = \{e\}.$$

Therefore

$$gh = hg.$$

Theorem 4.4.2. If K is the internal direct product of G and H, then

$$\boxed{K \cong G \times H.}$$

The isomorphism is

$$G \times H \to K, \qquad (g, h) \mapsto gh.$$

4.5 Why Direct Products Are Too Restrictive

The direct product assumes that the two factors do not interact.
In particular,

$$(g, e)(e, h) = (e, h)(g, e).$$

But many groups contain a normal subgroup N and another subgroup H such that

$$N \cap H = \{e\}$$

and

$$NH = G,$$

while elements of N and H do not commute.
In that case, the group is not a direct product.
Nevertheless, conjugation by elements of H may preserve N:

$$hnh^{-1} \in N.$$

This means that H acts on N by automorphisms.
The semidirect product records precisely this interaction.

4.6 Group Actions by Automorphisms

Let H and N be groups.

Definition 4.6.1. An action of H on N by automorphisms is a group homomorphism

$$\varphi : H \to \mathrm{Aut}(N).$$

For each $h \in H$,

$$\varphi(h) : N \to N$$

is an automorphism.
We often write

$$h \cdot n = \varphi(h)(n).$$

The action satisfies

$$e_H \cdot n = n$$

and

$$(h_1 h_2) \cdot n = h_1 \cdot (h_2 \cdot n).$$

Since each $\varphi(h)$ is an automorphism,

$$h \cdot (n_1 n_2) = (h \cdot n_1)(h \cdot n_2).$$

4.7 The Semidirect Product

Let

$$\varphi : H \to \mathrm{Aut}(N)$$

be an action.

Definition 4.7.1. The semidirect product of N by H with respect to φ is the set

$$N \rtimes_\varphi H = N \times H$$

with multiplication

$$\boxed{(n_1, h_1)(n_2, h_2) = (n_1 \varphi(h_1)(n_2), h_1 h_2)\,.}$$

The underlying set is the same as the Cartesian product, but the multiplication is twisted by the action.
The identity is

$$(e_N, e_H).$$

The inverse is

$$(n,h)^{-1} = \left(\varphi(h^{-1})(n^{-1}), h^{-1}\right).$$

Thus

$$\boxed{\text{semidirect product} = \text{direct product with a twisted multiplication.}}$$

4.8 Direct Products as a Special Case

Suppose the action is trivial:

$$\varphi(h) = \mathrm{id}_N$$

for every $h \in H$.
Then

$$(n_1, h_1)(n_2, h_2) = (n_1 n_2, h_1 h_2).$$

Therefore

$$\boxed{N \rtimes_\varphi H = N \times H}$$

when the action is trivial.
Thus direct products are special cases of semidirect products.
The distinction is exactly whether the second factor acts nontrivially on the first.

4.9 The Two Natural Subgroups

Inside

$$G = N \rtimes_\varphi H,$$

consider

$$\widetilde{N} = \{(n, e_H) : n \in N\}$$

and

$$\widetilde{H} = \{(e_N, h) : h \in H\}.$$

Then

$$\widetilde{N} \cong N$$

and

$$\widetilde{H} \cong H.$$

Moreover,

$$\widetilde{N} \trianglelefteq G,$$

while $\widetilde{H}$ need not be normal.
Also,

$$\widetilde{N} \cap \widetilde{H} = \{e\}$$

and

$$G = \widetilde{N}\widetilde{H}.$$

Hence the asymmetry of the notation

$$N \rtimes H$$

is intentional: N is the normal factor.

4.10 Conjugation Recovers the Action

The action φ can be seen directly inside the semidirect product.
We have

$$(e,h)(n,e)(e,h)^{-1} = (\varphi(h)(n), e).$$

Thus conjugation by the H-factor acts on N according to φ.
This is the structural meaning of the semidirect product:

$$\boxed{\text{external action} \varphi : H \to \mathrm{Aut}(N)}$$

is converted into

$$\boxed{\text{internal conjugation inside } N \rtimes_\varphi H.}$$

4.11 Internal Semidirect Products

The semidirect-product structure can also be recognized inside an existing group.

Theorem 4.11.1. Let G be a group containing subgroups N and H such that

$$N \trianglelefteq G,$$

$$N \cap H = \{e\},$$

and

$$NH = G.$$

Then

$$\boxed{G \cong N \rtimes_\varphi H,}$$

where

$$\varphi(h)(n) = hnh^{-1}.$$

Proof. Since N is normal,

$$hnh^{-1} \in N,$$

so conjugation defines

$$\varphi : H \to \mathrm{Aut}(N).$$

Define

$$F : N \rtimes_\varphi H \to G$$

by

$$F(n,h) = nh.$$

The condition

$$NH = G$$

gives surjectivity.
Suppose

$$nh = e.$$

Then

$$n = h^{-1}$$

belongs to both N and H. Hence

$$n = h = e$$

because

$$N \cap H = \{e\}.$$

Thus F is injective.
Finally, the twisted multiplication is precisely what is needed to make F a homomorphism.
□

4.12 Example: The Dihedral Group

Let

$$C_n = \langle r : r^n = e \rangle$$

be the cyclic group of order n.
Let

$$C_2 = \langle s : s^2 = e \rangle.$$

Define an action of C_2 on C_n by

$$srs^{-1} = r^{-1}.$$

Equivalently,

$$\varphi(s)(r) = r^{-1}.$$

Then the dihedral group is

$$\boxed{D_{2n} \cong C_n \rtimes C_2.}$$

Its defining relations may be written

$$r^n = e, \qquad s^2 = e, \qquad srs^{-1} = r^{-1}.$$

The group is generally not a direct product because

$$sr \neq rs.$$

Instead,

$$sr = r^{-1}s.$$

Thus the nontrivial action records the reflection reversing the direction of rotation.

4.13 Example: The Symmetric Group S_3

The alternating subgroup

$$A_3 = \{e, (123), (132)\}$$

is normal in S_3 and satisfies

$$A_3 \cong C_3.$$

Choose

$$H = \langle (12) \rangle \cong C_2.$$

Then

$$A_3 \cap H = \{e\}$$

and

$$A_3 H = S_3.$$

Conjugation by (12) sends

$$(123)$$

to

$$(132) = (123)^{-1}.$$

Therefore

$$\boxed{S_3 \cong C_3 \rtimes C_2.}$$

Again, the action of C_2 on C_3 is inversion.

4.14 Example: Affine Transformations

Consider transformations of the real line of the form

$$x \mapsto ax + b, \qquad a \neq 0.$$

Translations form a group

$$N = (\mathbb{R}, +),$$

while nonzero scalings form

$$H = (\mathbb{R}^\times, \cdot).$$

A scaling acts on a translation parameter by

$$a \cdot b = ab.$$

Therefore the affine group of the line has the structure

$$\boxed{\mathbb{R} \rtimes \mathbb{R}^\times.}$$

The multiplication law is

$$(b_1, a_1)(b_2, a_2) = (b_1 + a_1 b_2, a_1 a_2).$$

Indeed,

$$x \xmapsto{(b_2, a_2)} a_2 x + b_2$$

followed by

$$x \xmapsto{(b_1, a_1)} a_1 x + b_1$$

gives

$$a_1 a_2 x + (b_1 + a_1 b_2).$$

This is a concrete example of the twisted multiplication formula.

4.15 Semidirect Products and Split Extensions

Semidirect products are closely connected to the split short exact sequences introduced in Chapter 2.

Suppose

$$G = N \rtimes_\varphi H.$$

Define

$$i : N \to G, \qquad i(n) = (n, e),$$

and

$$\pi : G \to H, \qquad \pi(n, h) = h.$$

Then

$$1 \longrightarrow N \xrightarrow{i} G \xrightarrow{\pi} H \longrightarrow 1$$

is exact.
Moreover, define

$$s : H \to G$$

by

$$s(h) = (e, h).$$

Then

$$\pi \circ s = \mathrm{id}_H .$$

Therefore the sequence splits.
Thus

$$\boxed{N \rtimes H \Longrightarrow \text{split extension of } H \text{ by } N.}$$

4.16 From Split Extensions to Semidirect Products

Conversely, suppose

$$1 \longrightarrow N \xrightarrow{i} G \xrightarrow{\pi} H \longrightarrow 1$$

is a short exact sequence of groups and suppose there exists a homomorphism

$$s : H \to G$$

such that

$$\pi \circ s = \mathrm{id}_H .$$

Then $s(H)$ is a subgroup of G, and

$$G = i(N)s(H).$$

Moreover,

$$i(N) \cap s(H) = \{e\}.$$

Since $i(N)$ is normal in G, conjugation by $s(H)$ defines an action

$$\varphi : H \to \mathrm{Aut}(N).$$

Therefore

$$\boxed{G \cong N \rtimes_{\varphi} H.}$$

Hence:

$$\boxed{\text{split group extension} \longleftrightarrow \text{semidirect product.}}$$

This is the group analogue of the splitting discussion for modules, although the conclusion is different from an ordinary direct sum.

4.17 Direct versus Semidirect Products

It is useful to compare the two constructions directly.
For a direct product,

$$G = N \times H,$$

both factors are normal:

$$N \trianglelefteq G, \qquad H \trianglelefteq G.$$

Moreover, they commute:

$$nh = hn.$$

For a semidirect product,

$$G = N \rtimes H,$$

only N is required to be normal.
The factor H acts on N by conjugation:

$$hnh^{-1} = \varphi(h)(n).$$

Therefore:

$$\boxed{\text{direct product} = \text{semidirect product with trivial action.}}$$

4.18 The Role of Automorphism Groups

The possible semidirect products of N by H depend on homomorphisms

$$\varphi : H \to \mathrm{Aut}(N).$$

Thus understanding

$$\mathrm{Aut}(N)$$

is essential.
If

$$\mathrm{Aut}(N)$$

contains no nontrivial image of H, then every semidirect product with these factors is necessarily a direct product.
If several non-equivalent actions exist, several non-isomorphic semidirect products may arise.
Hence the data

$$N, \qquad H$$

alone do not determine the semidirect product.
The action

$$\varphi$$

is part of the structure.

4.19 Example with Trivial and Nontrivial Actions

Consider

$$N = C_3, \qquad H = C_2.$$

There is the trivial action

$$C_2 \to \mathrm{Aut}(C_3),$$

which gives

$$C_3 \times C_2 \cong C_6.$$

But

$$\mathrm{Aut}(C_3) \cong C_2,$$

so there is also a nontrivial action in which the generator of C_2 acts by inversion.
This gives

$$C_3 \rtimes C_2 \cong S_3.$$

Thus the same two component groups can produce different groups:

$$\boxed{C_6 \qquad \text{and} \qquad S_3.}$$

The difference lies entirely in the action.

4.20 Products as Universal Constructions

The original course notes also describe the direct product through projection maps and its mapping property. :contentReference[oaicite:1]index=1
For objects A and B, the product comes with maps

$$\pi_A : A \times B \to A, \qquad \pi_B : A \times B \to B.$$

Given another object X with maps

$$f : X \to A$$

and

$$g : X \to B,$$

there is a unique map

$$h : X \to A \times B$$

such that

$$\pi_A \circ h = f$$

and

$$\pi_B \circ h = g.$$

For sets, groups, modules, and vector spaces, this map is simply

$$h(x) = (f(x), g(x)).$$

Thus the direct product may be characterized by a universal property, not merely by an explicit Cartesian-product construction.

4.21 Direct Sums and Direct Products of Modules

For finitely many modules,

$$M_1, \ldots, M_r,$$

the direct sum and direct product coincide as underlying modules:

$$M_1 \oplus \cdots \oplus M_r = M_1 \times \cdots \times M_r.$$

For an infinite family

$$\{M_i\}_{i\in I},$$

the distinction becomes important.
The direct product is

$$\prod_{i\in I} M_i = \{(m_i)_{i\in I} : m_i \in M_i\}.$$

The direct sum is

$$\bigoplus_{i\in I} M_i = \{(m_i)_{i\in I} : m_i = 0 \text{ for all but finitely many } i\}.$$

Therefore

$$\boxed{\bigoplus_{i\in I} M_i \subseteq \prod_{i\in I} M_i.}$$

They agree for finite I, but generally differ for infinite I.
This distinction also appeared earlier when dual spaces of infinite direct sums were mentioned.

4.22 Universal Property of the Direct Sum

For modules M and N, let

$$\iota_M : M \to M \oplus N$$

and

$$\iota_N : N \to M \oplus N$$

be the canonical inclusions.
Given homomorphisms

$$f : M \to X, \qquad g : N \to X,$$

there exists a unique homomorphism

$$F : M \oplus N \to X$$

such that

$$F \circ \iota_M = f$$

and

$$F \circ \iota_N = g.$$

Explicitly,

$$F(m, n) = f(m) + g(n).$$

Thus the direct sum is characterized by maps *out of* the summands, while the direct product is characterized by maps *into* the factors.
Schematically:

$$\boxed{\text{product} : X \to A,\ X \to B \Longrightarrow X \to A \times B,}$$

whereas

$$\boxed{\text{direct sum} : A \to X,\ B \to X \Longrightarrow A \oplus B \to X.}$$

4.23 Internal and External Constructions

Several constructions in this chapter admit both an internal and an external viewpoint. The external direct product starts with two separate groups

$$G \quad \text{and} \quad H$$

and constructs

$$G \times H.$$

The internal direct product starts with subgroups of a larger group and asks whether

$$K \cong G \times H.$$

Similarly, the external semidirect product starts with

$$N, \qquad H, \qquad \varphi : H \to \mathrm{Aut}(N)$$

and constructs

$$N \rtimes_{\varphi} H.$$

The internal version begins with

$$N, H \leq G$$

and recognizes G as a semidirect product when

$$N \trianglelefteq G, \qquad N \cap H = \{e\}, \qquad NH = G.$$

Thus:

$$\boxed{\text{external construction} \longleftrightarrow \text{internal decomposition.}}$$

4.24 Structural Summary

The simplest way to combine two algebraic objects is the direct product:

$$G \times H.$$

Its multiplication is componentwise, and the two factors interact trivially.
A nontrivial action

$$\varphi : H \to \mathrm{Aut}(N)$$

modifies the multiplication and produces

$$N \rtimes_{\varphi} H.$$

Hence

$$\boxed{\text{trivial action} \longrightarrow \text{direct product,}}$$

while

$$\boxed{\text{nontrivial action} \longrightarrow \text{semidirect product.}}$$

The construction is closely related to exact sequences:

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1.$$

When this sequence splits,

$$G$$

can be reconstructed as a semidirect product:

$$\boxed{G \cong N \rtimes H.}$$

Thus the main chain of this chapter is

$$\boxed{\text{direct products} \longrightarrow \text{actions} \longrightarrow \text{semidirect products} \longrightarrow \text{split extensions.}}$$

The theme is the same one that has appeared throughout this volume: larger algebraic structures can often be understood by identifying simpler components together with the information describing how those components interact.

Chapter 5

Further Algebraic Structures and Perspectives

5.1 From Linear Algebra to Structural Algebra

The preceding chapters moved progressively away from ordinary finite-dimensional linear algebra.
Vector spaces were generalized to modules:

$$\text{vector spaces over fields} \longrightarrow \text{modules over rings.}$$

Subspaces and quotient spaces became submodules and quotient modules. Kernels and quotients were organized through exact sequences. Bilinear and multilinear maps led to tensor products, and actions between algebraic objects led to semidirect products and extensions.
These constructions suggest a broader principle:

an algebraic object is determined not only by its elements, but also by its operations and the maps pr

This viewpoint is one of the transitions from linear algebra toward modern algebra.
The present chapter collects several structures and perspectives that lie near this boundary. It is intended primarily as a map for future study rather than a complete development of any one subject.

5.2 Rings Revisited

A ring provides the scalar structure underlying module theory.

Definition 5.2.1. A ring R is a set equipped with two operations

$$+ : R \times R \to R$$

and

$$\cdot : R \times R \to R$$

such that $(R, +)$ is an abelian group, multiplication is associative, and multiplication distributes over addition.

If

$$ab = ba$$

for all $a, b \in R$, then R is commutative.
If there exists

$$1_R \in R$$

such that

$$1_R a = a 1_R = a,$$

then R is a ring with identity.
The original course notes introduce rings, commutative rings, fields, and modules at the beginning of the algebraic development. :contentReference[oaicite:3]index=3

5.3 Units and Zero Divisors

Definition 5.3.1. An element $u \in R$ is a unit if there exists $v \in R$ such that

$$uv = vu = 1.$$

The set of units is commonly denoted by

$$R^\times.$$

For a field K,

$$K^\times = K \setminus \{0\}.$$

For matrix rings,

$$M_n(K)^\times = GL_n(K).$$

The original notes explicitly identify units as invertible elements and give fields, matrix rings, and polynomial rings as examples. :contentReference[oaicite:4]index=4

Definition 5.3.2. A nonzero element $a \in R$ is a zero divisor if there exists a nonzero $b \in R$ such that

$$ab = 0$$

or

$$ba = 0.$$

The absence of zero divisors is one of the properties that makes a commutative ring resemble a field.

5.4 Integral Domains

Definition 5.4.1. A commutative ring R with identity is an integral domain if

$$ab = 0$$

implies

$$a = 0 \qquad \text{or} \qquad b = 0.$$

Every field is an integral domain.
The ring

$$\mathbb{Z}$$

is an integral domain but not a field.
If K is a field, then

$$K[X]$$

is also an integral domain.
Indeed, for nonzero polynomials,

$$\deg(fg) = \deg f + \deg g,$$

so

$$fg \neq 0.$$

This fact is recorded explicitly in the original notes. :contentReference[oaicite:5]index=5

5.5 Euclidean Domains

The division algorithm for integers and polynomials suggests a common structure.

Definition 5.5.1. An integral domain R is a Euclidean domain if there exists a function

$$N : R \setminus \{0\} \to \mathbb{N}$$

such that for every $a, b \in R$ with $b \neq 0$, there exist $q, r \in R$ satisfying

$$a = bq + r$$

with either

$$r = 0$$

or

$$N(r) < N(b).$$

For

$$R = \mathbb{Z},$$

one may take

$$N(a) = |a|.$$

For

$$R = K[X],$$

one may take

$$N(f) = \deg f.$$

The course notes explicitly use these two examples in discussing Euclidean-type division. :contentReference[oaicite:6]index=6

5.6 Principal Ideal Domains

Definition 5.6.1. A principal ideal domain, or PID, is an integral domain in which every ideal is generated by one element.

Thus every ideal $I \subseteq R$ has the form

$$I = (a) = \{ra : r \in R\}.$$

Important examples are

$$\mathbb{Z}$$

and

$$K[X].$$

The original notes specifically identify polynomial rings over a field as principal ideal domains in the structural summary of the early weeks. :contentReference[oaicite:7]index=7 The hierarchy

$$\boxed{\text{field} \Longrightarrow \text{Euclidean domain} \Longrightarrow \text{PID} \Longrightarrow \text{integral domain}}$$

provides increasingly general settings in which parts of familiar linear-algebraic reasoning can still survive.

5.7 Why $K[X]$ Appears in Linear Algebra

The polynomial ring

$$K[X]$$

is not merely an isolated example of a ring.
Let

$$T : V \to V$$

be a linear operator.
For

$$p(X) \in K[X],$$

define

$$p(X) \cdot v = p(T)v.$$

This gives V the structure of a $K[X]$-module.
Indeed,

$$(p + q)(T)v = p(T)v + q(T)v,$$

and

$$(pq)(T)v = p(T)q(T)v.$$

Therefore:

$$\boxed{\text{linear operator } T \quad \Longleftrightarrow \quad K[X]\text{-module structure on } V.}$$

This is one of the most important conceptual bridges between linear algebra and module theory.

5.8 Invariant Subspaces as Submodules

Under the $K[X]$-module structure

$$p(X) \cdot v = p(T)v,$$

a vector subspace

$$W \subseteq V$$

is a $K[X]$-submodule if and only if

$$T(W) \subseteq W.$$

Indeed, if W is T-invariant, then

$$T^k v \in W$$

for every $v \in W$, so

$$p(T)v \in W$$

for every polynomial p.
Thus

$$\boxed{T\text{-invariant subspaces} = K[X]\text{-submodules.}}$$

This explains why invariant-subspace theory and module theory are closely related.

5.9 Canonical Forms from the Module Viewpoint

In ordinary linear algebra, one studies the similarity class of

$$T : V \to V.$$

The $K[X]$-module viewpoint replaces this by the classification of the corresponding $K[X]$-module V.
Since

$$K[X]$$

is a PID, finitely generated module theory provides powerful decomposition results.
From this perspective, canonical forms such as rational canonical form and Jordan normal form are not isolated matrix tricks. They reflect decomposition theorems for modules over

$$K[X].$$

Schematically,

$$\boxed{\text{module decomposition over } K[X] \longrightarrow \text{canonical form of } T.}$$

This perspective is conceptually important, but a full proof of the structure theorem for finitely generated modules over a PID lies beyond the scope of these notes.

5.10 Algebras over a Field

There is another way to add structure to a vector space.

Definition 5.10.1. A K-algebra is a K-vector space A equipped with a bilinear multiplication

$$A \times A \to A.$$

Thus an algebra combines:

$$\boxed{\text{vector-space structure} + \text{multiplication.}}$$

Examples include:

$$M_n(K),$$

$$K[X],$$

and

$$\mathrm{End}_K(V).$$

For matrices, multiplication is associative but generally not commutative.
For $K[X]$, multiplication is both associative and commutative.
Thus different classes of algebras arise by imposing different identities on the multiplication.

5.11 Associative and Commutative Algebras

An algebra A is associative if

$$(ab)c = a(bc)$$

for every $a, b, c \in A$.
It is commutative if

$$ab = ba.$$

Hence

$$M_n(K)$$

is an associative algebra, while

$$K[X]$$

is a commutative associative algebra.
The study of commutative rings and commutative algebras develops into the subject known as commutative algebra.
Only its position in the larger algebraic landscape is recorded here; the original course notes available for these notes do not contain a systematic development of commutative algebra.

5.12 Ideals

Ideals play for rings a role analogous to submodules.

Definition 5.12.1. Let R be a commutative ring. An ideal $I \subseteq R$ is an additive subgroup satisfying

$$ri \in I$$

for every $r \in R$ and $i \in I$.

Equivalently, when R is regarded as an R-module over itself,

$$\boxed{\text{ideals of } R = R\text{-submodules of } R.}$$

Thus quotient rings

$$R/I$$

are simultaneously examples of quotient modules.
This is another example of module language unifying constructions that first appear in different contexts.

5.13 Prime and Maximal Ideals

The following definitions belong to the basic language of commutative algebra.

Definition 5.13.1. A proper ideal

$$\mathfrak{p} \subsetneq R$$

is prime if

$$ab \in \mathfrak{p}$$

implies

$$a \in \mathfrak{p} \qquad \text{or} \qquad b \in \mathfrak{p}.$$

Definition 5.13.2. A proper ideal

$$\mathfrak{m} \subsetneq R$$

is maximal if there is no ideal I satisfying

$$\mathfrak{m} \subsetneq I \subsetneq R.$$

These notions are characterized by quotients:

$$\boxed{\mathfrak{p} \text{ prime} \iff R/\mathfrak{p} \text{ is an integral domain,}}$$

and

$$\boxed{\mathfrak{m} \text{ maximal} \iff R/\mathfrak{m} \text{ is a field.}}$$

Every maximal ideal in a commutative ring with identity is prime.
These definitions are included only as orientation for later commutative algebra, not as a claim that the subject was developed systematically in the original course.

5.14 Bilinear Operations as Additional Structure

The course notes summarize several objects as a "K-module with extra structure." In particular, bilinear, symmetric, symplectic, quadratic, and sesquilinear structures are listed together. :contentReference[oaicite:8]index=8
This viewpoint is worth preserving.
A vector space begins only with

$$+, \qquad \text{scalar multiplication.}$$

Additional maps may then be imposed:

$$V \times V \to K,$$

$$V \times V \to V,$$

or more generally

$$V^{\times r} \to W.$$

Different identities satisfied by these maps produce different algebraic theories.
For example:

$$\text{symmetric form} \longrightarrow \text{quadratic geometry},$$

while

$$\text{alternating nondegenerate form} \longrightarrow \text{symplectic linear algebra}.$$

5.15 Symplectic Vector Spaces

A symplectic vector space provides one important example of a vector space with extra bilinear structure.

Definition 5.15.1. A symplectic form on a finite-dimensional vector space V is a bilinear form

$$\omega : V \times V \to K$$

that is alternating and nondegenerate.

Thus

$$\omega(v, v) = 0$$

for every $v \in V$, and

$$\omega(v, w) = 0 \quad \text{for every } w \in V$$

implies

$$v = 0.$$

A vector space equipped with such a form is called a symplectic vector space.
The original notes explicitly list symplectic forms among the additional structures placed on a K-module. :contentReference[oaicite:9]index=9

5.16 The Standard Symplectic Matrix

On

$$K^{2n},$$

define

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Then

$$J^T = -J$$

and

$$\det J \neq 0.$$

The bilinear form

$$\omega(x, y) = x^T J y$$

is alternating and nondegenerate.
Thus

$$(K^{2n}, \omega)$$

is the standard symplectic vector space.
A consequence is that a finite-dimensional symplectic vector space must have even dimension.

5.17 Symplectic Transformations

A linear transformation

$$T : V \to V$$

is symplectic if it preserves the symplectic form:

$$\omega(Tx, Ty) = \omega(x, y).$$

In standard coordinates this condition becomes

$$\boxed{T^T J T = J.}$$

The set of such matrices forms the symplectic group

$$Sp(2n, K).$$

This resembles the orthogonal condition

$$Q^T Q = I,$$

but the preserved bilinear form is alternating rather than symmetric.
Thus:

$$\boxed{\text{orthogonal geometry} : Q^T I Q = I,}$$

whereas

$$\boxed{\text{symplectic geometry} : S^T J S = J.}$$

5.18 Hamiltonian Matrices

The symplectic form naturally leads to another matrix class.

Definition 5.18.1. A matrix

$$H \in M_{2n}(K)$$

is Hamiltonian if

$$JH$$

is symmetric.

Equivalently,

$$\boxed{H^T J + JH = 0.}$$

Such matrices occur as infinitesimal generators of symplectic transformations.
Indeed, if

$$S(t) = e^{tH},$$

then the condition

$$H^T J + JH = 0$$

implies

$$S(t)^T J S(t) = J.$$

This gives the infinitesimal relation

$$\boxed{\text{Hamiltonian matrices} \longrightarrow \text{symplectic transformations.}}$$

This section records the basic algebraic bridge only; a systematic development of Hamiltonian systems or symplectic geometry belongs to later courses.

5.19 Lie Algebras as a Further Direction

Another way to place additional structure on a vector space is to introduce a bilinear bracket.
The following material is included as a guide to later study. The available course notes do not provide a systematic Lie algebra development.

Definition 5.19.1. A Lie algebra over a field K is a vector space

$$\mathfrak{g}$$

equipped with a bilinear map

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

such that

$$[x, x] = 0$$

for all $x \in \mathfrak{g}$, and the Jacobi identity

$$\boxed{[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0}$$

holds for all $x, y, z \in \mathfrak{g}$.

When the characteristic is not 2,

$$[x, y] = -[y, x].$$

Thus a Lie algebra is a vector space with an alternating bilinear operation satisfying an additional compatibility identity.

5.20 Matrix Lie Algebras

Every associative algebra produces a Lie algebra by defining

$$[A, B] = AB - BA.$$

In particular,

$$M_n(K)$$

becomes a Lie algebra under the commutator bracket

$$[A, B] = AB - BA.$$

This Lie algebra is denoted by

$$\mathfrak{gl}_n(K).$$

Several important subspaces are closed under this bracket.
For example,

$$\mathfrak{sl}_n(K) = \{A \in M_n(K) : \operatorname{tr}(A) = 0\}.$$

Indeed,

$$\operatorname{tr}(AB - BA) = 0,$$

so

$$[A, B] \in \mathfrak{sl}_n(K)$$

whenever $A, B \in \mathfrak{sl}_n(K)$.

5.21 Lie Groups and Infinitesimal Structure

Matrix groups studied in linear algebra often possess corresponding Lie algebras.
For example, the orthogonal group satisfies

$$Q^T Q = I.$$

Consider a differentiable curve

$$Q(t)$$

with

$$Q(0) = I.$$

Differentiating

$$Q(t)^T Q(t) = I$$

at $t = 0$ gives

$$X^T + X = 0,$$

where

$$X = Q'(0).$$

Thus the infinitesimal objects associated with the orthogonal group are skew-symmetric matrices.
Similarly, differentiating

$$S(t)^T J S(t) = J$$

gives

$$X^T J + J X = 0.$$

This is precisely the Hamiltonian matrix condition.
Hence

$$\boxed{\text{global transformation group} \longrightarrow \text{infinitesimal Lie algebra.}}$$

This relationship explains why Lie algebra language appears naturally near advanced linear algebra.

5.22 Categories and Morphisms

The original notes also begin using category-theoretic language, including objects, morphisms, monomorphisms, epimorphisms, and universal properties. :contentReference[oaicite:10]index=1
At its most elementary level, a category consists of objects and maps between them.
Examples include:

$$\mathbf{Set}, \qquad \mathbf{Grp}, \qquad R\text{-}\mathbf{Mod}, \qquad K\text{-}\mathbf{Vect}.$$

The precise internal nature of the objects changes, but the language of maps, composition, and identity maps remains.
This makes it possible to describe constructions such as products, direct sums, tensor products, and quotients by their mapping properties rather than by particular coordinate formulas.

5.23 Monomorphisms and Epimorphisms

In categorical language, injectivity and surjectivity are replaced by properties defined entirely through composition.

Definition 5.23.1. A morphism

$$f : A \to B$$

is a monomorphism if

$$f \circ g_1 = f \circ g_2$$

implies

$$g_1 = g_2.$$

Definition 5.23.2. A morphism

$$f : A \to B$$

is an epimorphism if

$$g_1 \circ f = g_2 \circ f$$

implies

$$g_1 = g_2.$$

For vector spaces and modules,

$$\text{monomorphism} \iff \text{injective homomorphism},$$

and

$$\text{epimorphism} \iff \text{surjective homomorphism}.$$

The course notes explicitly use this language while reviewing modules and universal properties. :contentReference[oaicite:11]index=11

5.24 Universal Properties as a Common Language

Several constructions in this volume were defined through universal properties.
The tensor product is characterized by

$$\text{bilinear maps from } M \times N$$

factoring uniquely through

$$M \otimes N.$$

The direct product is characterized by maps into its factors.
The direct sum is characterized by maps out of its summands.
The quotient is characterized by maps that vanish on a chosen submodule.
These definitions appear different in explicit coordinates, but they share a common pattern:

specify the maps an object should support $\longrightarrow$ characterize the object uniquely up to isomorphism.

This is one of the conceptual reasons universal properties become central in more advanced algebra.

5.25 A Map of the Subjects

The structures encountered in this volume point toward several later subjects.
Module theory leads naturally toward

$$\text{homological algebra}, \qquad \text{representation theory},$$

and

$$\text{commutative algebra}.$$

Tensor products lead toward

$$\text{multilinear algebra}, \qquad \text{differential geometry}, \qquad \text{representation theory}.$$

Symplectic forms and Hamiltonian matrices lead toward

$$\text{symplectic geometry}$$

and

$$\text{Hamiltonian mechanics}.$$

Lie brackets and matrix groups lead toward

Lie theory.

Ideals and modules over commutative rings lead toward

commutative algebra

and eventually

algebraic geometry.

These arrows should not be interpreted as dependencies in a single course. They are directions in which the elementary constructions of this volume reappear.

5.26 What Has Been Learned and What Has Been Recorded

The first two volumes of these notes were organized as a core undergraduate theory of advanced linear algebra.
This volume has a deliberately different status.
Its central constructions are

modules,	quotients and exact sequences,	tensor products,	actions and extensions.

The final topics of this chapter—commutative algebra, symplectic structures, Hamiltonian matrices, Lie algebras, and categorical language—serve primarily as signposts.
They indicate where the structural language developed earlier begins to connect with larger mathematical theories.
Accordingly, the purpose of this volume is not to claim mastery of every subject that appears in it.
Its role is

to preserve the frontier reached by the course.

Definitions and basic constructions are recorded so that unfamiliar ideas can be located again when they reappear with the necessary background in later study.

5.27 The Three-Volume Perspective

The complete organization of these notes can now be summarized.
The first volume moves from computation to structure:

systems $\longrightarrow$ matrices $\longrightarrow$ vector spaces $\longrightarrow$ linear operators $\longrightarrow$ diagonalization.

The second volume develops finer finite-dimensional structure:

Jordan form $\longrightarrow$ inner products $\longrightarrow$ spectral theory $\longrightarrow$ quadratic forms $\longrightarrow$ SVD.

The present volume moves beyond ordinary linear algebra:

modules $\longrightarrow$ exact sequences $\longrightarrow$ tensor products $\longrightarrow$ actions and extensions $\longrightarrow$ further algebraic str

The common theme is the gradual replacement of coordinates by structure.
At the beginning, matrices are the main objects of calculation.
At the next level, matrices represent linear maps.

Beyond that, vector spaces themselves become examples of modules, bilinear maps become linear maps on tensor products, and canonical constructions are characterized through universal properties.
Thus the conceptual progression of the three volumes may be written as

$$\boxed{\text{calculation} \longrightarrow \text{linear structure} \longrightarrow \text{canonical structure} \longrightarrow \text{abstract algebraic structure.}}$$

This is the point at which the present notes stop. The theories indicated here are not endings of advanced linear algebra, but beginnings of later mathematics.

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This volume records the more advanced material encountered in the course, including modules, exact sequences, tensor products, semidirect products, and several further algebraic structures. Some of these topics remain subjects for future study rather than material I would claim to have fully mastered. Their inclusion here is intended to preserve the mathematical frontier reached in the course and to provide a reference for returning to these ideas when they arise again in later study.